◆ AI PMO · INTELLIGENCE LAYER

SAP PS · P6 / MS Project · AI synthesis

AI PMO

BUSINESS EDITION

How It Works: Capabilities, Formulas,
and the Reasoning Behind Them

An AI-assisted PMO intelligence layer · 2026

Table of Contents

Foreword

This is the Business Edition of the AI PMO book. It is written for the people who run projects and portfolios — PMO leads, project-controls and commercial managers, and the executives they report to. Its promise is simple: for every capability AI PMO provides, you will understand *what question it answers*, the *formula in plain terms*, *why that method is the right one* (the standard behind it), and a *worked example*. There is almost no code here; its companion, the Technical Edition, covers how everything is built.

The book follows a deliberate arc. It opens with the problem AI PMO exists to solve — the seam between the cost system and the schedule — and the discipline that keeps the product in its lane. It then walks the **money story** end to end: earned value, forecasting, revenue recognition, cash flow, margin, and change. From there it turns to risk, issues and performance; to how the intelligence is made legible; and finally to the standards and governance that keep every number defensible.

A note on the figures: worked examples use a synthetic demonstration portfolio, not any real project or organisation.

As of June 2026 the system described here is no longer a prototype on a developer machine: it runs live — a Next.js application on Railway over a cloud Supabase database — and every screenshot in this edition is captured from the published application.

How to read this book

Each capability has a Business chapter here and a counterpart in the Technical Edition; cross-references connect the two (for example, A5 ↔ B12). Chapters are self-contained, but the money-story chapters (A5–A10) are best read in order, as each builds on the last.

Part I — Why AI PMO Exists

The seam between the cost system and the schedule — and the discipline of building only what no other system owns.

The Problem AI PMO Solves

Why “are we healthy?” is the one question neither system can answer

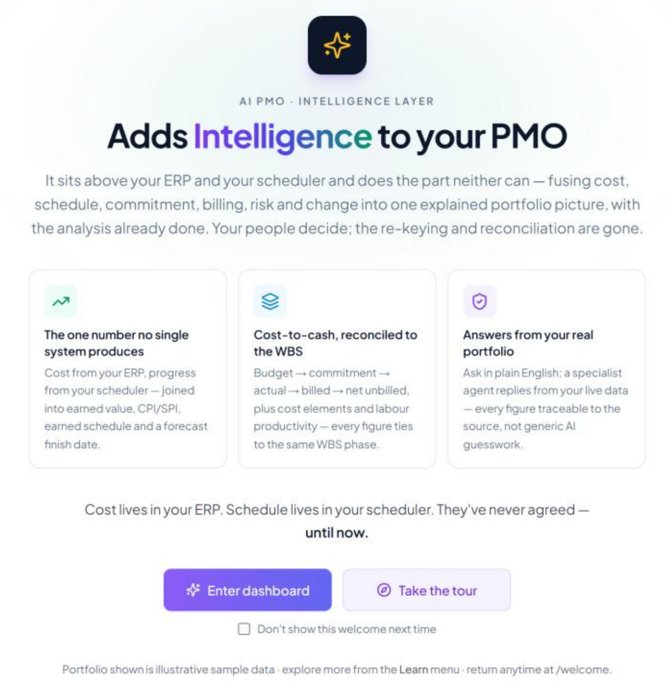


Figure 1 — The welcome gateway — what AI PMO is, in one screen.

Every large EPC contractor runs its projects on two authoritative systems that do not talk to each other in the way the work demands — and the questions that decide a project's fate all fall in the gap between them.

On one side sits the **ERP — SAP Project System (PS)**: the system of record for the **commercials**. The WBS, the budget, commitments and purchase orders, actual cost, billing, revenue recognition. It knows, to the cent, what the project has cost and what it is owed.

On the other sits the **scheduler — Primavera P6 or Microsoft Project**: the system of record for **time**. The activity network, the logic, the critical path, physical progress. It knows when things happen and how far along they are.

Both are excellent at their jobs. Neither, on its own, can tell you whether the project is actually *healthy* — because health lives in the **combination**.

The seam is where the answers live

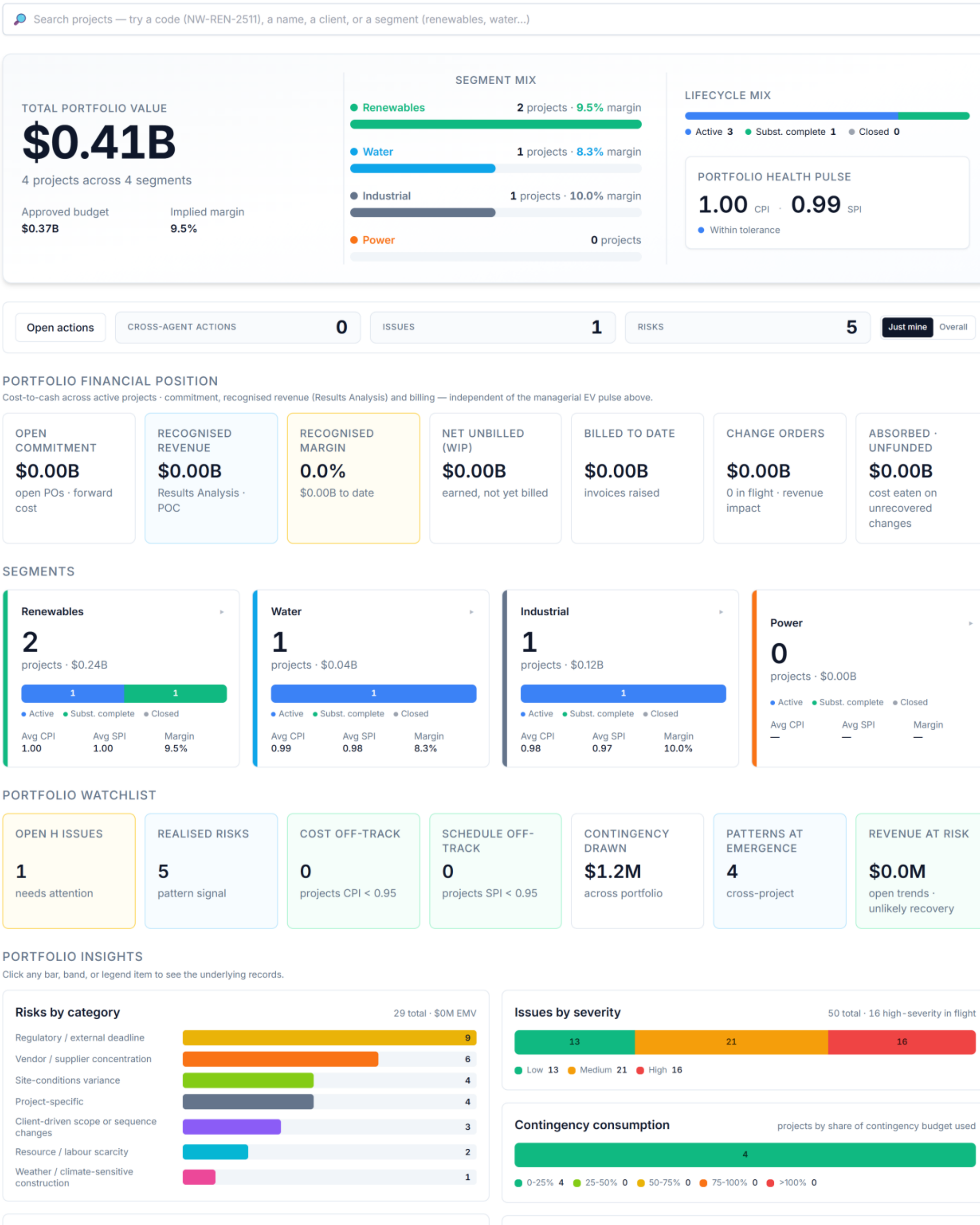


Figure 2 — The executive portfolio dashboard — AI PMO's synthesis across SAP PS and the scheduler.

The questions executives and project-controls leads actually ask all sit across the seam between cost and time:

- *Are we getting value for the money we've spent?* — needs cost (SAP) **and** progress (scheduler).
- *What will this finish at, and when?* — needs the cost run-rate **and** the schedule trajectory.
- *How much revenue can we book, and how much cash will it tie up?* — needs contract and cost **and** billing timing.
- *Is our risk and change exposure eroding the margin we sold?* — needs the commercial baseline **and** the live registers.

None of these can be answered inside either system alone. In most organisations they are answered — slowly, monthly — by an analyst exporting both systems into a spreadsheet, reconciling them by hand, and writing a narrative. That spreadsheet *is* the missing layer: fragile, late, person-dependent, and invisible to everyone but its author.

What AI PMO is

AI PMO is that missing layer, built properly: an **AI-assisted PMO intelligence layer** that sits across the SAP PS ↔ scheduler seam, joins the two systems on the **WBS code**, and **synthesises** the answers — earned value, forecasts, revenue recognition, cash flow, margin bridges, risk and change exposure — with a plain-language narrative and recommendations on top.

It does not replace either system of record. It reads their authoritative data and produces the decision-grade synthesis neither can produce alone. It is not an ERP and not a scheduler: it does not own the cost ledger, run the critical-path engine, or store transactional truth. A capability belongs in AI PMO only if it is **synthesised** from a system of record or read from a named source. The moment it would need to *own* transactional data, it has left its lane.

Why now

Two things make this layer buildable today that weren't a few years ago. The data in SAP PS and the schedulers is now reliably accessible through modern APIs, and large language models can turn reconciled numbers into the narrative and recommendations that used to require a senior analyst. AI PMO is what you get when you point those two capabilities at the seam every EPC business already feels but few have systematised.

> **The truth about a project lives in the seam between cost and time** — the one place neither system of record can see, and the one place AI PMO is built to stand.

Part II — The Money Story

Six chapters that follow a contract dollar end to end: earned value, forecast, revenue, cash, margin, change.

Earned Value, in Plain Terms

The one measure that tells you whether a project is really on track

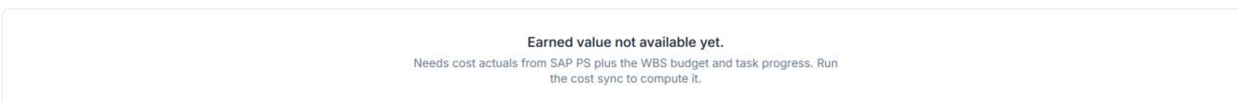


Figure 3 — Earned value on a project: the S-curve (PV/EV/AC), CPI/SPI, earned schedule, and EV by WBS branch.

"Are we on track?" is usually answered with a half-truth — *"we've spent 60% of the budget"* (which says nothing about how much work that bought) or *"we're 60% done"* (which says nothing about what it cost). Earned value puts the two together into one honest verdict — and AI PMO produces it continuously, cost from your ERP and progress from your scheduler, with no spreadsheet reconciliation in between.

Three figures, one currency

It rests on three numbers, all expressed in budgeted dollars so they compare directly:

- **Planned Value (PV)** — the budgeted cost of the work you *planned* to have done by now. The baseline.
- **Earned Value (EV)** — the budgeted cost of the work you have *actually* done. This is the figure neither system holds on its own; it comes from pricing the scheduler's progress at the ERP's budget.
- **Actual Cost (AC)** — what that work actually cost, straight from the ERP.

Earned value is the pivot: it turns "how much work" into "how much that work was worth" — the thing that finally lets cost and schedule be judged on one scale.

Two indices a leader reads in seconds

- **Cost — CPI = EV ÷ AC.** *"For every dollar spent, how many dollars of work did we get?"* Above 1.0, under budget; below, over.
- **Schedule — SPI = EV ÷ PV.** *"How much of the planned work have we actually done?"* Above 1.0, ahead of plan; below, behind.

A reading of **CPI 0.92, SPI 0.96** is an instant verdict — meaningfully over cost, slightly behind — the answer a manual reconciliation would take a week to reach.

Honest to the finish, and precise about where

- **Earned schedule.** The plain schedule index has a known flaw: near the end of a job it drifts back to "on plan" even when the project finishes late. AI PMO also reports schedule performance in *weeks ahead or behind*, so the timing read stays truthful right through completion.
- **Down to the branch.** A single project-level number hides as much as it reveals. AI PMO breaks earned value down the work-breakdown structure, so *"the project is at 0.92"* becomes *"engineering is fine; procurement is carrying the overrun"* — a headline turned into a place to act. The same view rolls up across the whole portfolio.

What you do with it

The reading above — 8% over cost, 4% behind — is more a cost problem than a schedule one, so the response points at productivity and rates rather than resequencing, and the branch breakdown says where to look first. It is the foundation the rest of the money story builds on — the forecast extrapolates it, variance analysis explains its gaps, and revenue recognition deliberately stands apart from it.

> **Earned value is the one measure that cannot be flattered by looking at cost or schedule alone** — the difference between *reporting activity* and *managing performance*.

Worked figures use the synthetic demonstration portfolio.

Forecasting Cost & Revenue to EAC

What will this job finish at, what will we be paid, and what margin will we land?

Every month a project must answer one question honestly: *given what we now know, what will this cost to finish, what will we be paid, and what margin will we land?* The **Estimate at Complete (EAC)** is the cost half of that answer, recognised revenue is the income half, and the difference is the forecast margin. Doing this well, on a fixed cadence, is the single most important discipline in project controls — because the absolute number matters less than **how it moves, and why.**

AI PMO produces the answer continuously, drawing cost and progress from SAP and the scheduler joined on the WBS, so the forecast is seen *moving* rather than reconstructed by hand each month.

The core identity

EAC is not a calculation, it is a **re-forecast**:

> **EAC = actual cost to date (AC) + estimate to complete (ETC)**

The actuals are a fact from the ledger. All the judgement sits in the **ETC** — the forward look. The monthly job is to refresh the actuals and the progress, re-estimate the ETC, and read off the EAC, the variance at complete (VAC = BAC − EAC), and the trend versus last month.

Forecasting the ETC

Two registers of method work together:

- **Bottom-up re-estimate — the authoritative forecast.** Cost engineers re-estimate the remaining work package by package: remaining quantities at current rates, open commitments still to run off, and an allowance for scope not yet committed. This is the EAC of record because it reflects what the delivery team actually knows.
- **Earned-value formulaic EAC — the check, not the forecast.** Three quick extrapolations bracket the bottom-up number:
  - `EAC = BAC ÷ CPI` — today's cost efficiency continues (typical),
  - `EAC = AC + (BAC - EV)` — the remainder runs at plan (optimistic),
  - `EAC = AC + (BAC - EV) ÷ (CPI × SPI)` — both cost and schedule pressure persist (pessimistic).

If the bottom-up EAC falls outside that bracket, someone is being optimistic — the signal to challenge it. The reality test is the **To-Complete Performance Index**:

TCPI = (BAC − EV) ÷ (EAC − AC)

— the cost efficiency the *remaining* work must achieve to hit the chosen EAC. If the TCPI sits well above the run-rate CPI, the forecast is not credible and the EAC should move up.

The monthly cycle

A disciplined period close runs in order:

1. **Cut-off and capture.** Freeze a date; pull actuals, open commitments, and physical progress (percent complete, installed quantities, timesheets).
2. **Update earned value.** EV = percent complete × BAC, per WBS.
3. **Re-forecast the ETC.** Bottom-up on the live and material packages, formulaic elsewhere; fold in open trends, absorbed cost, escalation, and contingency drawdown.

4. **Roll up the cost forecast.** EAC = AC + ETC; VAC = BAC − EAC; run the TCPI sanity check.
5. **Forecast revenue.**
6. **Margin and bridge.** Forecast margin = forecast revenue − EAC; reconcile as-sold → as-built.
7. **Variance / trend review.** The heart of the close: explain *why the EAC and recognised revenue moved this month versus last*. That movement, with causes, is the real output — not the absolute figure.
8. **Sign-off.** The monthly cost review challenges and approves the forecast.

The revenue forecast

On lump-sum EPC, revenue is recognised **over time by percentage of completion**, and it follows the cost forecast rather than running independently:

- **POC** is cost-based — `AC ÷ EAC` — or measured from physical progress.
- **Recognised revenue = POC × (contract value + approved variations + probable claims)**, with claims constrained under IFRS 15 until recovery is *highly probable*.
- **Forecast revenue at complete** = contract value plus approved and probable changes — exactly the funded / at-risk split a trend register produces.

Each month the *incremental* revenue is recognised and reconciled to billing; the gap is work in progress (under-billed) or deferred revenue (over-billed).

One physical-progress spine

The crucial discipline is that cost and revenue are **not** forecast independently. They reconcile through the same physical progress:

progress → earned value → EV/EAC → cost-based POC → recognised revenue → revenue − EAC = forecast margin.

If the cost POC and the revenue POC run off different progress, margin drifts and the forecast loses integrity. One spine, two readings.

The control output: EAC movement

Because the absolute EAC matters less than its motion, the monthly deliverable is the **EAC movement** — the change versus the prior close, decomposed into its causes. The simplest useful split separates the period's EAC change into:

- a **scope** component — budget growth from approved changes, and
- a **cost-performance** component — the residual run-rate efficiency shift.

A rising EAC driven by *scope* is recoverable (it should be matched by a contract increase); a rising EAC driven by *performance* is margin erosion and demands action. Naming which is which, every month, is what turns a forecast into a control.

AI PMO keeps a **month-end forecast snapshot** per project — BAC, EV, AC, the re-forecast EAC / ETC / VAC and CPI / SPI, plus the revenue side (contract value, cost-based POC, recognised revenue, billed, forecast margin). From that series it derives, with no extra entry, a cost-and-revenue trend plotted across the closes, the EAC-movement decomposition for the latest close, and a reconciliation back to the trend register (funded changes lift the contract, absorbed cost lifts the EAC) and the margin bridge (as-sold → as-built). The same snapshots roll up to the portfolio, so *"is our forecast getting better or worse, and why?"* can be answered for every project at once.

> **A forecast that only updates the absolute EAC hides the story** — decomposing the monthly movement surfaces margin erosion while there is still time to act.

Revenue Recognition

Of this contract, how much have we actually earned — and therefore how much profit can we book?

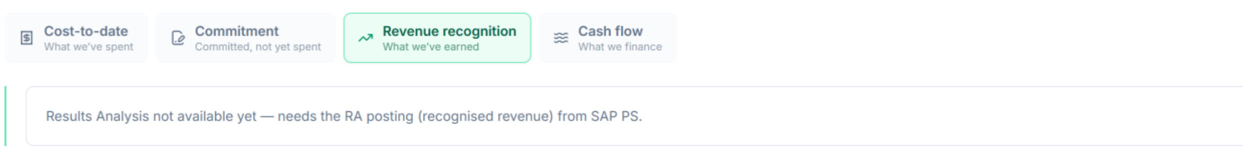


Figure 4 — The Revenue-recognition lens: cost-based POC, recognised revenue, cost of sales, margin, and WIP/deferred.

A project can have spent a great deal of money, invoiced the client for a great deal more, and still not be able to put either number in the profit-and-loss account as revenue. **Revenue recognition** answers a precise, audited question: *of this contract, how much revenue have we earned to date — and therefore how much profit can we book?* It turns project progress into a number an auditor will accept.

Why recognition is not earned value

This is the single most important idea here, and the reason recognition is treated apart from earned value rather than as a footnote to it.

Earned value is a *managerial* measure. It uses physical or schedule-based progress to tell a project manager whether the work is ahead or behind and over or under cost. Internal, fast, tuned for control.

Revenue recognition is a *financial* measure. It feeds the audited statements, so it must follow an accounting standard (IFRS 15), use a consistent and defensible basis, and reconcile to the general ledger. External, formal, tuned for assurance.

The two can legitimately give different "percent complete" figures for the same project on the same day, because they measure different things for different audiences. A project 60% complete by physical progress might recognise revenue at 71% on a cost basis. **Neither is wrong.** Treating them as interchangeable is the classic error AI PMO is built to avoid — which is why it labels the Results Analysis panel *"independent of the managerial earned value on the EV tab,"* and builds it from the posted Results Analysis figures rather than reusing the earned-value percentage.

The method: cost-based percentage-of-completion

Revenue is recognised **over time** using the **cost-based percentage-of-completion (POC)** method — the mechanism SAP PS runs as *Results Analysis*. AI PMO does not perform the recognition; it displays the figure SAP PS posts, then explains the method and reconciles it against billing. In plain terms: revenue is recognised in proportion to the cost incurred against the total cost expected.

Results Analysis applies the method phase by phase at the WBS level; AI PMO reads the posted figures and rolls them up:

1. **Percentage of completion.** For each phase, *POC = actual cost to date ÷ planned (or expected) cost*, capped at 100%. Purely cost-based, computed independently of earned value.
2. **Planned revenue.** Each phase carries a share of the total contract value, set by its planned margin and scaled so the phases sum to the full contract.
3. **Recognised revenue.** *Recognised = POC × planned revenue* — the revenue you may book for the phase to date.
4. **Cost of sales.** The actual cost incurred to earn that recognised revenue.
5. **Recognised margin.** *Margin = recognised revenue − cost of sales*, with margin percentage being margin ÷ recognised revenue.

Because recognition is driven by cost incurred, a phase that has spent 71% of its expected cost recognises 71% of its planned revenue — regardless of how the scheduler rates its physical progress.

Three numbers, three questions

On a mature project AI PMO holds three different "value earned / value billed" numbers at once, and the discipline is never to confuse them:

Figure	Basis	Question it answers
Earned value (EV $)	Physical / schedule progress × budget	How much work have we *done*?
Recognised revenue	Cost-based POC × contract (IFRS 15)	How much can we *book*?
Billed to date	Invoices raised against the contract	How much have we *asked the client to pay*?

These rarely match, and the *gaps between them* are themselves the insight.

WIP versus deferred revenue

Recognition and billing run on different clocks. You recognise revenue as you earn it; you bill it according to the contract's payment milestones. The difference lands on the balance sheet:

- **WIP — work in progress (a contract asset).** When you have *recognised more than you have billed*, the excess is revenue earned but not yet invoiced. The client owes it to you even though no invoice has gone out. *WIP = recognised revenue − billed* (when positive).
- **Deferred revenue (a contract liability).** When you have *billed more than you have recognised* — front-loaded or milestone billing ahead of the work — the excess is money invoiced for work not yet earned. You still owe the client the work. *Deferred = billed − recognised revenue* (when positive).

AI PMO computes both automatically, per phase and for the project, so a commercial lead can see at a glance whether the project is a net lender to the client (carrying WIP) or has been paid ahead (carrying deferred revenue) — a direct read on cash and on commercial risk.

Why IFRS 15

IFRS 15 ("Revenue from Contracts with Customers") is the governing standard, and the method above is its application to long-duration construction-type contracts. Revenue is recognised as the entity satisfies a **performance obligation**. For an EPC contract the obligation is typically satisfied **over time** — the asset is built on the customer's site, under the customer's control, as it progresses — so revenue is recognised progressively rather than all at completion. A cost-based input method — cost incurred relative to total expected cost — is an accepted way to measure that progress. Using a standard rather than an internal rule of thumb is what makes the number audit-ready: consistent across projects, defensible to a reviewer, and reconcilable to the ledger.

A worked example

Take a project mid-execution, as AI PMO's Results Analysis panel would show it:

- **Recognised revenue: $387.9M** — cost-based POC applied across the phases.
- **Cost of sales: $351.3M** — actual cost incurred to earn it.
- **Recognised margin: $36.6M (9%)** — recognised revenue minus cost of sales.
- **Billed to date: $440.7M** — invoices raised against the contract.
- **Deferred revenue: $52.8M** — because billed ($440.7M) exceeds recognised ($387.9M), the $52.8M difference is a *contract liability*: the project has been paid ahead of the work it has earned.

The story: the job is recognising a thin 9% margin so far, and it is **billed ahead of recognition by $52.8M**. That is healthy for cash — the client has funded work not yet earned — but it is a liability to discharge: that revenue can only be booked as the remaining cost is incurred. A controller reading this knows not to mistake the strong billing position for booked profit. Had the position been reversed, recognised exceeding billed, the same panel would show **WIP / unbilled** — a contract asset flagging revenue earned but not yet invoiced, and a prompt to get the billing out.

AI PMO **synthesises** this from data that lives in SAP; it does not own the ledger. It reads the posted Results Analysis figures per WBS phase (POC, recognised revenue, cost of sales, recognised margin), the contract value, and the billing events; rolls the phases up to a project position; and derives WIP versus deferred from recognised-minus-billed. The recognition POC stays cost-based and computed separately, so the managerial and financial views can never silently contaminate each other. Recognition policy, cut-off, and the formal posting remain finance's and SAP's responsibility — AI PMO's job is to make the earned-versus-billed-versus-recognised picture legible and connect it back to cost, schedule and change.

> **Recognised revenue is what you may book, not what you have done or what you have billed** — treating the three as one number is the classic error that turns a strong billing position into imaginary profit.

Worked figures use the synthetic demonstration portfolio.

Cash Flow & Funding Exposure

How much of our own money will this project tie up, when does funding peak, and when does it turn cash-positive?

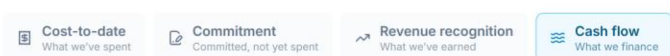


Figure 5 — The project cash-flow lens: cumulative cash in vs out, the funding gap, peak funding, and the cash-positive crossover.

Profit and cash are not the same thing, and on a long EPC contract they can move in opposite directions for months. A project can be recognising healthy margin and still be haemorrhaging cash — because it pays its suppliers and labour long before the client pays it. **Cash-flow forecasting** answers the treasury question margin never does: *how much of our own money will this project tie up, when does that funding requirement peak, and when does the project finally start paying for itself?*

Cash is not cost, revenue, or earned value

This is a distinct lens on the same project:

- **Cost** is what you have *committed and consumed*.
- **Recognised revenue** is what you may *book* — an accounting event.
- **Earned value** is *progress* against budget.

- **Cash flow** is *actual money moving in and out of the bank* — and it moves on the contract's **payment terms**, not on when work is done or revenue is recognised.

The defining feature of construction cash flow is **timing**. You incur cost now, pay it on your supplier terms, bill the client on milestones, and collect on their terms — typically later. The lag between paying out and collecting in is the whole story.

The two curves

AI PMO models a project's cash position as two cumulative curves over time, synthesised from data already in SAP and the forecast — no new ledger, no new entry:

Cash out — cumulative payments. The project's cost curve (actual cost to date, then ramping to the forecast cost at completion, EAC) **shifted later by the payment lag**, because you pay a cost some weeks after you incur it. The default payment lag is one month.

Cash in — cumulative collections. The project's billing curve (billed to date, then ramping toward the contract value) **shifted later by the collection lag** — the time between raising an invoice and the client's cash arriving. The default collection lag is two months, and the curve is taken **net of retention**.

Both curves ramp smoothly between the latest actuals and their end points, so the forecast portion is a believable glide rather than a straight line.

The funding gap is working capital you finance

The vertical distance between the two curves is the heart of the chart:

> **Net cash position = cumulative cash in − cumulative cash out.**

While cash out runs above cash in — which it does for most of an EPC project's life — the **net is negative**, and that shaded gap is **working capital the contractor is financing out of its own balance sheet or facilities.** It is real money with a real cost: every dollar of that gap is a dollar borrowed, or a dollar of the firm's cash not earning elsewhere. This is why a profitable project can still be a cash problem: margin tells you the job will end up ahead; the funding gap tells you how much you must carry to *get* there.

Peak funding and the cash-positive crossover

The single most important number on the chart is the **peak funding need** — the deepest point of the negative net, the maximum cash the project will ever have tied up at once, and *when* it occurs. This is the figure a treasurer sizes facilities against and a finance director wants on one line: *"this job will need up to $X of funding, worst around month Y."* AI PMO reads it straight off the trough of the net curve.

As the project winds down, billing and collections catch up to and overtake spending, the net curve climbs back through zero, and the project becomes **cash-positive** — from that month on it returns cash rather than consuming it. AI PMO reports the crossover month. Deep early funding, late recovery: the classic EPC cash curve, and seeing the crossover date lets a business plan when the project stops being a drain on group cash.

Retention

Construction contracts almost always withhold **retention** — a percentage of each payment (the model uses 5%) held back by the client until completion or the end of the defects period. It matters to cash because it is money you have earned and billed but will **not collect until the very end**. AI PMO models the collectible cash-in net of retention, so both the peak funding and the cash-positive point reflect the cash you can actually expect — not the headline billing.

A worked example

A project mid-build, as AI PMO's funding-exposure chart would show it:

- **Net cash position: −$77.6M** to date — the project is currently carrying $77.6M of working capital.
- **Peak funding need: −$91.7M, around March** — the deepest the hole gets; the firm must be able to fund roughly $92M at the worst point.
- **Cash-positive: June** — the forecast month the net crosses back above zero and the project starts returning cash.
- **Terms (assumed): collect 2 months / pay 1 month · 5% retention** — the levers behind the curves.

The story: this job consumes up to ~$92M of cash before it turns the corner in June. That is not a margin problem — it is a *funding* problem, and exactly the number that should be agreed with treasury before the project, not discovered during it. It also points straight at the levers: pull the collection lag in by a month, or de-front-load the retention, and the peak shrinks.

The portfolio view

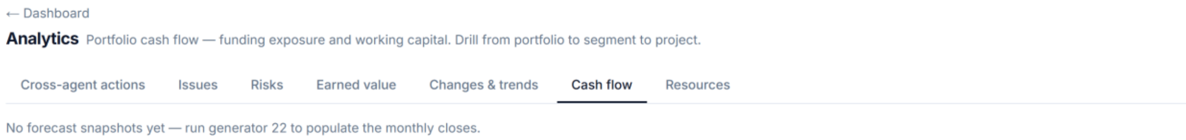


Figure 6 — Portfolio funding exposure, with drill-down to segment and project.

A single project's funding need is useful; the **portfolio's** is decisive, because peaks that are survivable one at a time can collide. AI PMO rolls every active project's cash curve onto one shared monthly timeline — a project contributes nothing before it starts and its final position after it finishes — to produce the whole book's funding shape: total working capital tied up, the portfolio peak funding requirement and when it lands, and the aggregate cash-positive crossover. It drills the same number down to segment and then to a single project, so a treasurer sees not just *how much* funding the portfolio needs but *which* projects and segments are driving the peak.

Because cash flow is pure synthesis — the two curves derived from the cost/EAC forecast and the billing/contract figures, each shifted by the lags and netted for retention — it updates the moment cost, billing, or the forecast move. The lags and retention are **explicit, adjustable assumptions**, shown on the chart, not hidden, so the forecast is honest about what it depends on. Actual cash positions, drawdowns and facility management stay with finance; AI PMO's job is to see the peak coming.

> **A profitable project can still be a cash problem** — margin says the job ends ahead, but the funding gap says how much of your own money you must carry to get there.

Worked figures use the synthetic demonstration portfolio.

Margin: As-Sold → As-Planned → As-Built

Where the profit you signed up for actually goes

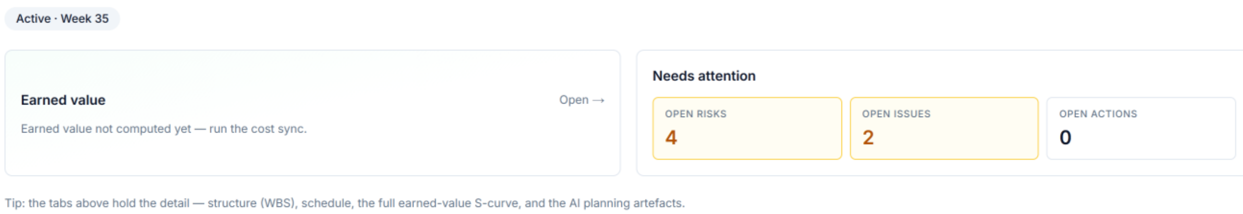


Figure 7 — The project Overview, where the margin bridge sits alongside earned value.

Every project is won on a margin — the profit the bid promised. By completion it has almost always become a *different* margin. The question worth answering is never just *"what's our margin now?"* but *"how did we get from the margin we sold to the margin we're forecasting, and what moved it?"* AI PMO draws that bridge: it decomposes the journey into named, attributable steps, so erosion is explained rather than merely observed.

Three margins, one project

AI PMO tracks the same project's margin at three points in its life:

- **As-sold** — the margin at booking: *sold contract value − the frozen baseline budget*. What the deal promised and what the business expects to keep.
- **As-planned** — the margin in the current plan: *current contract value − current WBS budget*. The team has detailed and re-estimated; the plan may have moved off the bid.
- **As-built** — the forecast margin at completion: *current contract value − estimate at completion (EAC)*. Where the job is actually heading, given performance to date.

As-sold is the promise, as-planned is the intention, as-built is the truth. The gaps between them are the story — and because each margin draws on a different source (the frozen booking baseline, the live plan, the EAC), they are never silently blended.

The bridge: what moved the margin

Reading left to right from as-sold to as-built, AI PMO attributes the change to named drivers — the same idea as a profit bridge in an annual report:

- **Budget growth** — where the as-planned budget grew above the as-sold baseline (re-estimation, scope detailing). Erodes margin if revenue didn't follow.

- **Change orders** — funded variations that move *both* revenue and cost; their net effect on margin can be accretive or dilutive.
- **Performance / forecast** — the move from as-planned to as-built, i.e. the cost over- or under-run the EAC now predicts against the plan.

Each step is a labelled increment in the waterfall, so a reviewer sees not just that margin fell from, say, 12% to 8%, but that *3 points went to budget growth on the baseline and 1 point to a cost overrun the forecast now carries, partly offset by an accretive change order*. That is a conversation about causes, not a number to argue about.

Why this matters

Margin erosion discovered at completion is a post-mortem; margin erosion *attributed in-flight* is a management lever. If the bridge shows the loss is budget growth on the baseline, that points to estimating discipline; if it is performance, that points to execution; if it is dilutive change, that points to commercial terms. The same headline number demands different responses depending on *which step* moved it — and only the bridge tells you which.

A worked example

Illustratively, a job booked at a 12% margin:

- **As-sold: 12%** — sold contract over the frozen baseline budget.
- **As-planned: 9%** — the baseline budget grew 3 points on re-estimation; revenue was unchanged, so margin fell.
- **As-built: 8%** — a cost overrun the EAC now forecasts costs a further point, partly offset by a small accretive change order.

The takeaway is specific: most of the erosion — 3 of the 4 points — happened at *planning*, where the bid was optimistic against the detailed estimate, not in the field. That sends the lesson upstream to estimating, where execution metrics alone would have wrongly blamed the site team.

> **Margin erosion attributed in-flight is a lever, not an autopsy** — because the same lost point demands a different response depending on which step moved it.

Worked figures use the synthetic demonstration portfolio.

Change & Trend Management

Catching margin movement while you can still do something about it

No change orders on this project to date.

Figure 8 — The project change & trend register.

On any sizeable contract, the scope delivered is never exactly the scope sold. Ground turns out different from the survey; the client asks for more; a design develops; an estimate proves light. Each of these moves cost — and sometimes revenue — away from the baseline. Capturing those movements, pricing them, and recovering what is recoverable is the difference between a project that holds its margin and one that quietly bleeds it.

The hard part is that most changes start life as a **grey area**. You rarely know at the moment of discovery whether the customer will pay. And under most contracts you **cannot stop work** while you find out — the notice-and-proceed obligation means you keep building, incurring cost, while the commercial outcome is still open. A model that only recognises a change once it has been formally agreed is therefore always behind reality.

The certainty continuum

The cleanest way to think about changes is as a single continuum of **certainty**, not as separate registers competing for the same record:

	Risk	Trend	Outcome
Nature	uncertain future event	emerging, near-certain cost movement	resolved change
Probability	< 100% (P × I)	≈ 100% (it is happening)	settled
Financial home	contingency (EMV)	cost forecast (EAC)	contract / margin
Resolves to	becomes a trend if it realises	a funded change order, or absorbed cost	—

A risk that materialises becomes a trend. A trend, once worked, resolves into either a **funded change order** (the customer agrees to pay) or an **absorbed (unfunded) change** (the work was done but there is no recovery, so it lands as a straight margin hit). These are not three competing instruments — they are three points along one lifecycle.

What a trend is

A **trend** is an emerging cost or schedule movement, logged the moment it is foreseen, before it has been formalised. The **trend register** is the ledger of those movements sitting between the budget baseline and the forecast at completion (EAC) — the mechanism that explains *how the forecast moved* between formal change orders. Every potential change enters here first; the register then tracks each one through to its outcome. Recognising cost when it is foreseen, not when the paperwork catches up, is what lets the forecast stay honest.

Why an unfunded change is not a risk

It is tempting to push unfunded cost growth — an estimating error, rework, a productivity shortfall — into the already-rich risk register. That is wrong once the movement is certain, for three concrete reasons:

1. **It distorts the risk numbers.** A discovered error has a probability of one; logging it as a risk corrupts the expected-monetary-value exposure, which is meant to be *expected*, not *actual*.
2. **It double-counts or misplaces the money.** Risk impact is covered by *contingency*; certain cost growth belongs in the *EAC*. Putting it in the risk register either double-counts it or draws down contingency for something it was never meant to cover.
3. **It breaks an audit boundary.** Cost controllers read the risk register and the cost forecast as distinct instruments; blending certain cost growth into risks undermines the contingency-drawdown discipline.

So while a cost movement is genuinely uncertain, it is a risk, managed via mitigation and contingency. Once it is discovered and near-certain, it leaves the risk register and becomes a trend.

Working at risk: the proceed obligation

Because work continues while the commercial position is unresolved, an open trend has a **split certainty**:

- The **cost is near-certain** — you are spending it regardless — so it goes into the EAC now.
- The **revenue is uncertain** — it is a claim that depends on the customer agreeing — so it is carried *at risk* until settled.

This is where risk-thinking legitimately re-enters: not on the cost, but on the **recovery**. Each open trend carries a **recovery confidence** — the probability that the customer funds it. From that, two honest figures follow for the open (in-negotiation) trends:

- **Expected recovery** = claimed revenue × recovery confidence.
- **Revenue at risk** = claimed revenue × (1 − recovery confidence).

Revenue at risk is the amount of cost already committed on unagreed changes that may never be recovered. It is the single most useful early-warning number in change management, and it is invisible to any tool that only records changes once they are executed.

The accounting gate: IFRS 15 variable consideration

The recovery-confidence model is not just a controls convenience — it mirrors the revenue-recognition rule. Under IFRS 15, a claim or unapproved variation is **variable consideration**, and revenue on it may not be recognised until recovery is *highly probable* of no significant reversal. Until that threshold is met, the cost is in the forecast but the revenue is **constrained**. A trend's recovery confidence is, in effect, the management view of that recognition gate: low confidence → constrained, no revenue taken; high confidence → recognisable.

Mapping to SAP

The split maps cleanly onto an SAP-centric estate, which is why it is faithful rather than abstract:

- A **funded change order** is an SD / contract change plus recognised PS revenue — the contract value grows.
- An **unfunded (absorbed) change** is **cost-only on the WBS** — a budget supplement with no sales-order change and no revenue.
- An **open trend** is forecast cost on the WBS with revenue held or blocked until the variation is approved.

The same data therefore reconciles to the SAP figures rather than living beside them.

The AI PMO model

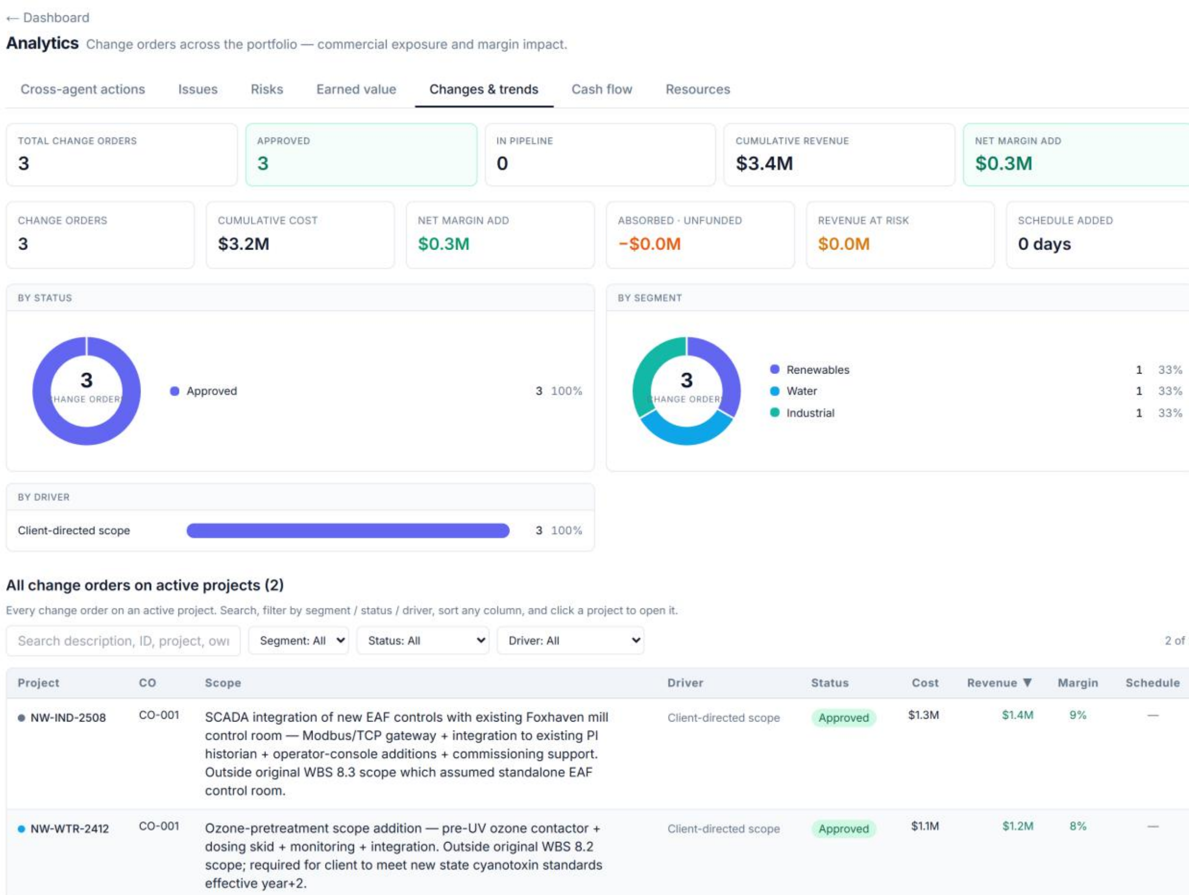


Figure 9 — Portfolio change & trend — funded vs absorbed, with revenue-at-risk.

AI PMO implements the continuum as a single **change & trend register**. Every entry carries a lifecycle status and a recovery confidence, and resolves to one of three outcomes:

> **Identified → Quantified → Submitted to client → In negotiation →** then **Approved (funded change order)** · **Absorbed (unfunded — margin hit)** · **Withdrawn (no impact)**

From that single model the platform derives, with no extra data entry:

- a **funded / absorbed / at-risk split** — recoverable revenue, absorbed cost (the margin hit), and revenue at risk with an average recovery confidence;
- a **status pipeline** from trend to outcome, with value at each stage;
- a **margin-impact** read — whether the change book is accretive or dilutive against the project's base margin, with the absorbed and dilutive entries flagged;
- a **by-driver** view (client-directed scope, site conditions, design development, estimating & productivity, regulatory, supply & escalation);
- a **portfolio roll-up** of the same, so exposure and revenue-at-risk are visible across every project at once.

A specialist change-order-reviewer agent reasons over this register — change exposure, pricing, and margin protection — and the whole thing reconciles to the margin bridge (funded variations move the contract step; unfunded cost growth moves the budget and execution steps) and to earned value (an absorbed cost shows up as negative cost variance).

The lineage we keep

A trend is the **middle** of a longer, conditional lifecycle — not the start of it:

> **Risk** (might happen) → *materialises* → **Issue** (has happened) **+ Trend** (the cost/schedule movement) → *worked* → **funded change order** (customer pays) **or absorbed, unfunded change** (margin hit).

The chain is conditional at every step: most risks never materialise, many issues are born directly with no prior risk, and a trend can arise on its own. Each register stands alone; the lifecycle is a set of optional hand-offs, never a forced pipeline. AI PMO records the hand-offs where they happen, so a commercial lead can trace a change back to its origin — a change or trend carries a *"Traces to R-009 → I-021"* chip. Lineage is captured, never enforced.

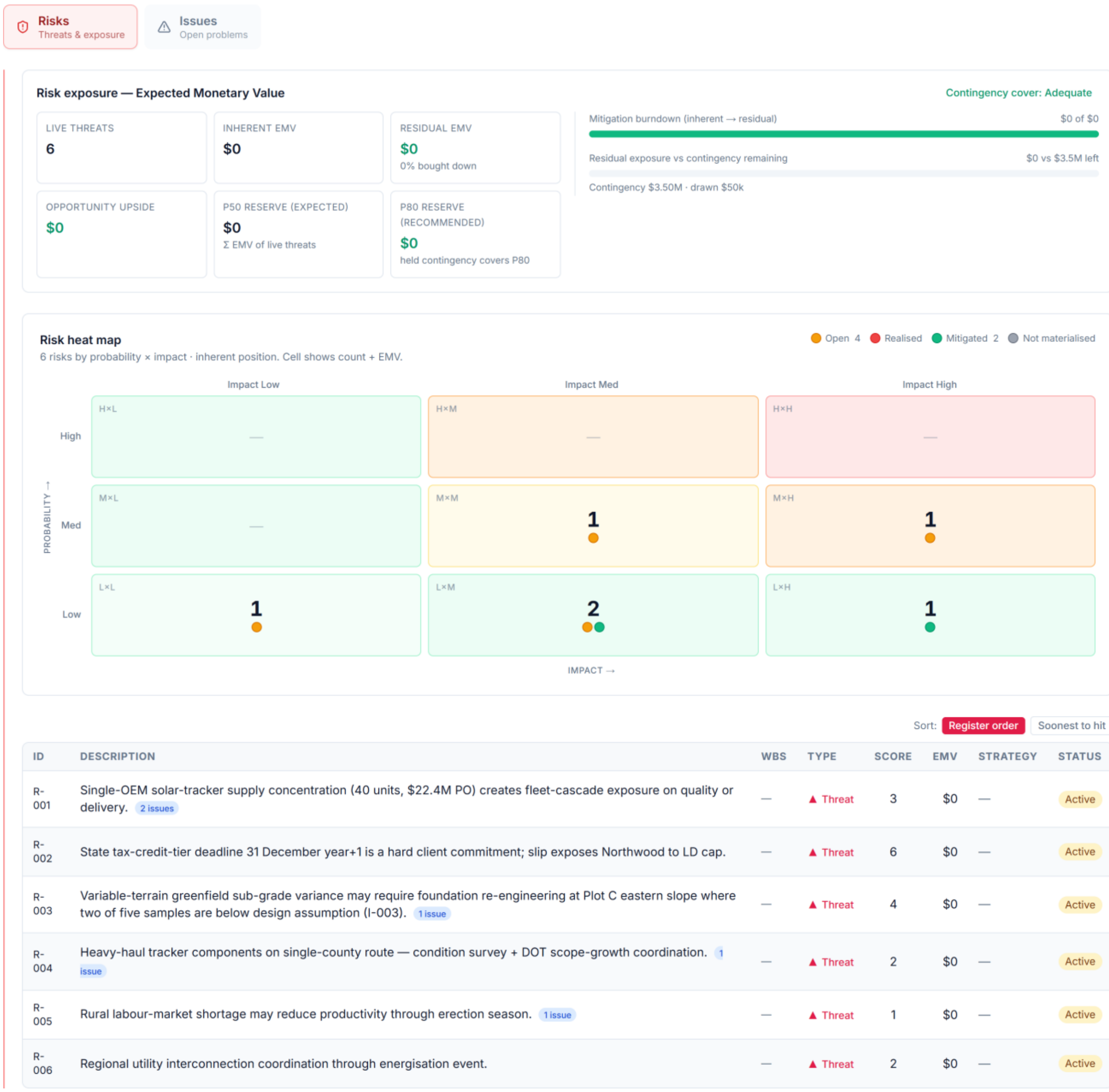
> **Absorbed cost is the silent killer of EPC margin** — making it a named, quantified category instead of an unexplained gap in the forecast is what lets a team see it early and act.

Part III — Risk, Issues & Performance

Quantified risk, disciplined issue management, and the weekly variance rhythm that keeps forecasts honest.

Risk: Expected Value, Residual Exposure & Contingency

What is this project's risk worth — and does the buffer cover it?



ID	DESCRIPTION	WBS	TYPE	SCORE	EMV	STRATEGY	STATUS
R-001	Single-OEM solar-tracker supply concentration (40 units, $22.4M PO) creates fleet-cascade exposure on quality or delivery. 2 issues	—	▲ Threat	3	$0	—	Active
R-002	State tax-credit-tier deadline 31 December year+1 is a hard client commitment; slip exposes Northwood to LD cap.	—	▲ Threat	6	$0	—	Active
R-003	Variable-terrain greenfield sub-grade variance may require foundation re-engineering at Plot C eastern slope where two of five samples are below design assumption (I-003). 1 issue	—	▲ Threat	4	$0	—	Active
R-004	Heavy-haul tracker components on single-county route — condition survey + DOT scope-growth coordination. 1 issue	—	▲ Threat	2	$0	—	Active
R-005	Rural labour-market shortage may reduce productivity through erection season. 1 issue	—	▲ Threat	1	$0	—	Active
R-006	Regional utility interconnection coordination through energisation event.	—	▲ Threat	2	$0	—	Active

Figure 10 — The project risk register: inherent vs residual EMV, with mitigation.

A risk register that only says *"high / medium / low"* cannot answer the question a finance director actually asks: *how much money is this project's risk worth, has our mitigation actually bought it down, and is our contingency enough to cover what's left?* AI PMO makes risk quantitative and connects it straight to the contingency buffer — pricing the register the team maintains rather than running the workshop that fills it.

Expected Monetary Value

The basic unit is **Expected Monetary Value (EMV)** — *probability × impact*. A threat with a 30% chance of a $10M overrun carries an EMV of $3M. Summing EMV across the register turns a list of worries into a single, comparable number: the risk-weighted cost the project is carrying. Opportunities (favourable risks) carry a positive EMV — potential upside — and are tracked separately from threats.

Inherent versus residual: did mitigation work?

The same risk has two EMVs:

- **Inherent EMV** — *before* mitigation: the raw probability × impact.
- **Residual EMV** — *after* mitigation: the reduced probability (and/or impact) once the agreed actions are in place.

The gap between them is the **reduction** — *(inherent − residual) ÷ inherent* — and it is the single most useful management number here, because it quantifies what mitigation has actually achieved. A register where inherent and residual are equal is a register where nothing has been *done*; a large reduction is mitigation earning its keep. AI PMO rolls this up to a portfolio exposure: total inherent, total residual, and the percentage bought down.

Only *live* risks — open or actively managed — count toward residual exposure. Risks that have not materialised carry no remaining exposure, and ones that *have* materialised have become issues or realised cost.

Is the contingency enough?

Quantified residual exposure lets AI PMO do what a colour-coded register cannot: test the **adequacy of contingency.** Coverage is *remaining contingency ÷ residual exposure*, and it sorts projects into bands — **Adequate**, **Tight**, **Exposed** — so a portfolio lead can see at a glance which projects are under-buffered for the risk they still carry. The buffer becomes a calculated position measured against the exposure that remains, not a round-number guess.

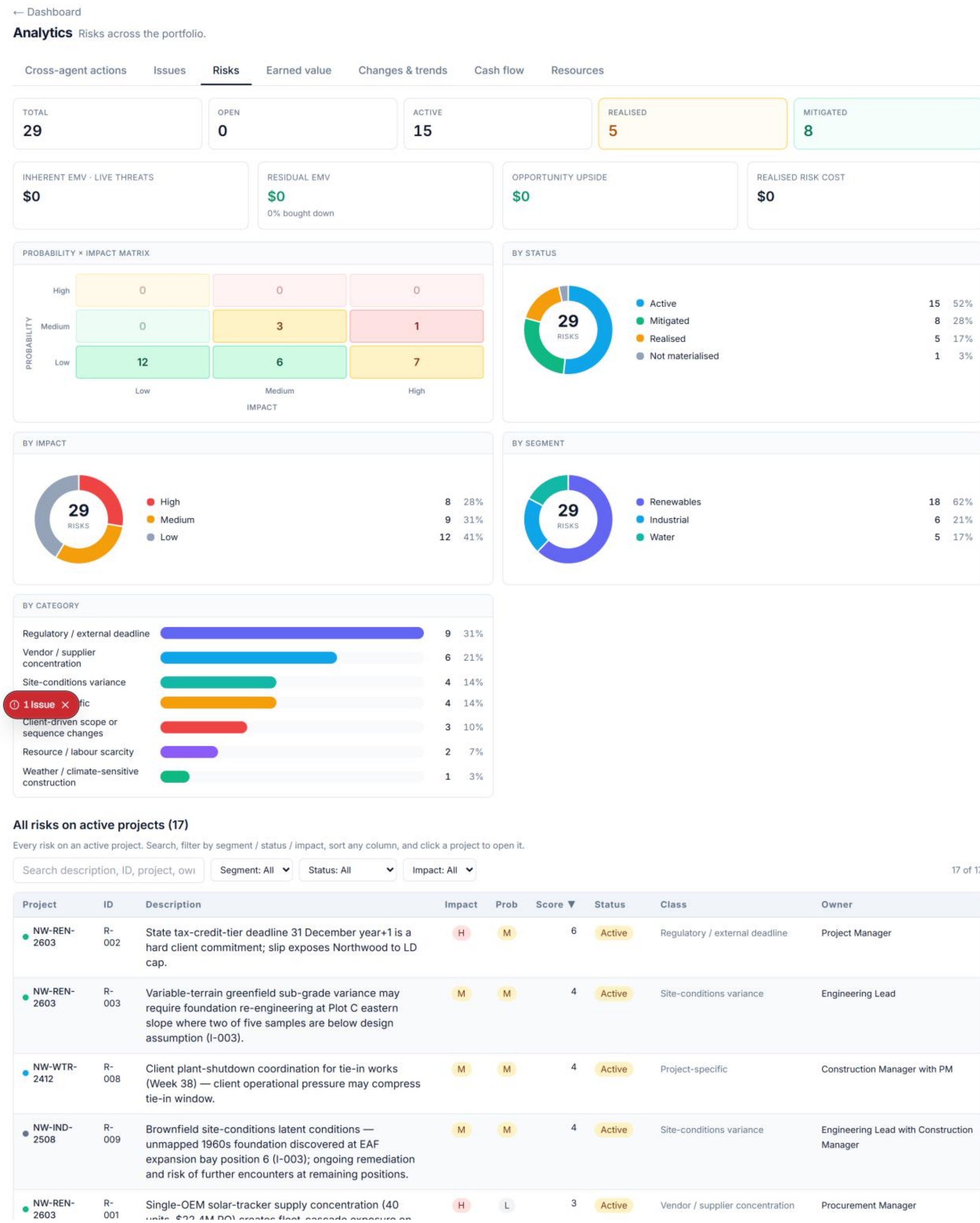


Figure 11 — Portfolio risk analytics — EMV exposure and the risk heatmap.

Why this matters

Three things a qualitative register can't give you, all here: a **portfolio risk price** you can compare and trend, **evidence that mitigation is working** (the buy-down), and an **early warning when the buffer is thin** for the exposure that remains. Risk stops being a compliance artefact and becomes a number that participates in the cost and contingency conversation.

A worked example

Illustratively, a project's register:

- **Inherent EMV: $14M** — raw risk-weighted exposure.
- **Residual EMV: $9M** — after mitigation; a **36% reduction** — mitigation is working, but $9M of live exposure remains.
- **Remaining contingency: $7M** → **coverage 0.78** → **"Tight."** The buffer sits below the residual exposure; the project is under-provisioned for the risk it still carries and should either draw down risk further or top up contingency.

> **A priced register turns risk into a number the buffer can be judged against** — the difference between a compliance colour-code and a contingency you can defend.

Worked figures use the synthetic demonstration portfolio.

Issue Management: Aging, SLA, Priority & Escalation

Turning an issue log into a managed queue

Most issue logs are a flat list that grows until someone panics. The questions that matter — *which issues are overdue, which are about to be, what should I work on first, and what needs to go up the chain right now?* — go unanswered because the log carries no notion of time or urgency. AI PMO turns the log into a managed queue with aging, service levels, priority and escalation, all derived from the log the team already keeps.

A risk that happened

An **issue is a risk that has materialised.** Risk management is about things that *might* happen; issue management is about things that *have*. When a live risk triggers, it leaves the register and enters the issue log — which is why issues carry cost and schedule impact, and why a healthy process tracks the hand-off rather than letting risks quietly disappear.

Aging and SLAs

Every open issue has an **age** — how long since it was raised — and a **service level (SLA)**: the number of weeks it *should* take to resolve, defaulted by severity (a High issue gets a tighter SLA than a Low). Comparing the two sorts every issue into an aging band:

- **On track** — comfortably within SLA.
- **At risk** — past 75% of its SLA; about to breach.
- **Overdue** — past SLA; breached.

This single derivation changes the log from *“here is everything”* to *“here is what is slipping”* — the only view a manager can act on.

Priority: what to work on first

Not all open issues deserve equal attention. AI PMO scores **priority = severity × age (capped)** — so a high-severity issue that has been festering rises to the top, while a fresh low-severity one sits below it. The queue sorts itself by genuine urgency rather than by whoever shouted last.

Escalation: what needs to go up now

Some issues shouldn’t wait for the next review. AI PMO flags an issue for **escalation** when it is open, high-severity, and either already overdue *or* has no owner — and hasn’t been escalated already. That rule catches exactly the two failure modes that hurt: serious issues breaching their SLA, and serious issues that nobody owns. Surfacing them automatically is the difference between escalation-by-process and escalation-by-accident.

Resolution speed (MTTR)

Looking backwards, **mean time to resolve (MTTR)** — the average weeks from raised to closed — measures how fast the team actually clears issues. Trended, it shows whether the project is keeping pace or falling behind, and it is the honest counterpart to the open-queue view: a small open log with a worsening MTTR is not the good news it appears to be.

Why this matters

Aging plus SLA gives **early warning** before a breach; priority gives a **defensible work order**; escalation gives a **safety net** for the serious and the orphaned; MTTR gives a **throughput trend.** Together they convert an inert list into a queue that tells the team what to do next and tells management what is at risk.

A worked example

Illustratively, a project with 40 open issues:

- **6 overdue, 5 at risk** — eleven need attention now or imminently, not forty.
- **Top of the priority queue:** a high-severity interface issue, 9 weeks old against a 4-week SLA — high severity × large age puts it first.
- **3 flagged for escalation** — high-severity and overdue or unowned; these go up this week regardless of the review cycle.
- **MTTR trending from 3.1 → 3.8 weeks** — resolution is slowing even as the open count looks stable; a leading sign the team is losing ground.

The full lifecycle — and why it is not a pipeline

Risk, issue and change are three points on one continuum, but the path between them is *conditional*, not a fixed sequence:

> **Risk** (it *might* happen) → *materialises* → **Issue** (it *has* happened) **and a Trend** (the cost/schedule movement it creates) → *worked* → a **funded change order** (the customer pays) **or** an **absorbed, unfunded change** (a straight margin hit).

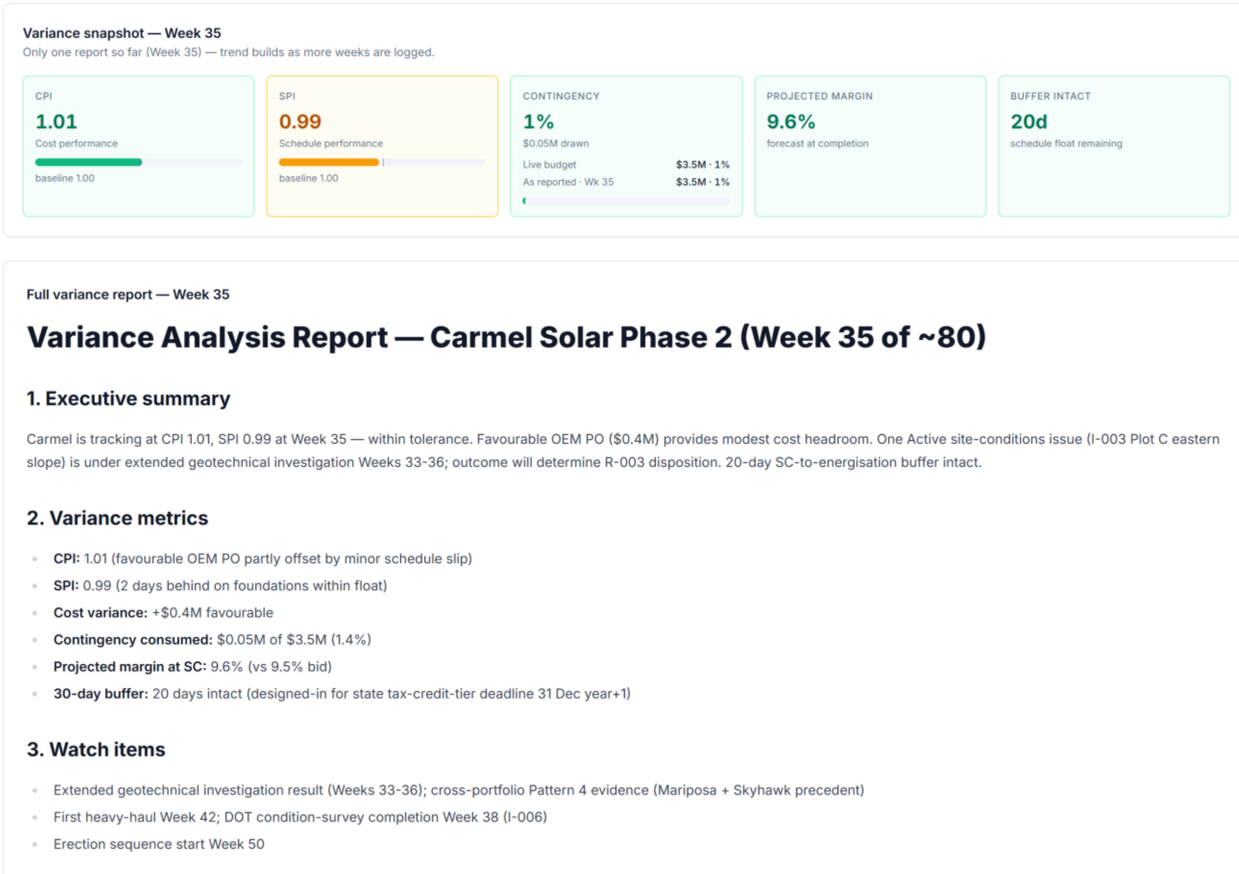
Every step is optional. Most risks never materialise; many issues are **born directly**, with no prior risk; and some trends arise on their own. So the registers each stand alone — the chain exists only where a real hand-off happened, and a database that *enforced* it would be wrong. AI PMO captures that lineage without enforcing it: an issue points back to its risk (shown as *"Materialised from risk"*), and a change or trend points back to the issue or risk it came from (a *"Traces to …"* chip).

> **Aging, priority and escalation turn an inert list into a queue that tells the team what to do next** — and tells management what is at risk before it breaches.

Worked figures use the synthetic demonstration portfolio.

Variance Analysis: CV, SV, VAC & the TCPI Reality Test

How far off are we — and is recovery still realistic?


Variance snapshot — Week 35
Only one report so far (Week 35) — trend builds as more weeks are logged.

CPI	SPI	CONTINGENCY	PROJECTED MARGIN	BUFFER INTACT
1.01	0.99	1%	9.6%	20d
Cost performance	Schedule performance	$0.05M drawn	forecast at completion	schedule float remaining
baseline 1.00	baseline 1.00	Live budget $3.5M · 1% As reported · Wk 35 $3.5M · 1%		

Full variance report — Week 35

Variance Analysis Report — Carmel Solar Phase 2 (Week 35 of ~80)

1. Executive summary

Carmel is tracking at CPI 1.01, SPI 0.99 at Week 35 — within tolerance. Favourable OEM PO ($0.4M) provides modest cost headroom. One Active site-conditions issue (I-003 Plot C eastern slope) is under extended geotechnical investigation Weeks 33-36; outcome will determine R-003 disposition. 20-day SC-to-energisation buffer intact.

2. Variance metrics

- **CPI:** 1.01 (favourable OEM PO partly offset by minor schedule slip)
- **SPI:** 0.99 (2 days behind on foundations within float)
- **Cost variance:** +$0.4M favourable
- **Contingency consumed:** $0.05M of $3.5M (1.4%)
- **Projected margin at SC:** 9.6% (vs 9.5% bid)
- **30-day buffer:** 20 days intact (designed-in for state tax-credit-tier deadline 31 Dec year+1)

3. Watch items

- Extended geotechnical investigation result (Weeks 33-36); cross-portfolio Pattern 4 evidence (Mariposa + Skyhawk precedent)
- First heavy-haul Week 42; DOT condition-survey completion Week 38 (I-006)
- Erection sequence start Week 50

Figure 12 — Project variance analysis.

Indices like CPI and SPI tell you a project is off track. Variance analysis turns that into the questions a leader actually has to answer: *how many dollars and weeks are we off, how far off will we finish, and — the one most reports avoid — can we still recover?* AI PMO reads the gaps and applies the honesty test.

Where we are, in money

Two variances locate the project against its plan, both stated in dollars so they carry a size, not an adjective:

- **Cost variance (CV) = EV — AC.** Earned value minus actual cost. Negative means you have spent more than the work is worth — over cost.
- **Schedule variance (SV) = EV — PV.** Earned value minus planned value. Negative means you have accomplished less than planned by now — behind schedule.

Expressed as a figure — and as a percentage of work done — these say not "behind" but *"$8M of work behind."*

Where we'll end up

Variance at completion (VAC) = BAC − EAC — the budget at completion minus the forecast at completion. It is the bottom line: the total over- or under-run the project is heading for if nothing changes, with VAC% (VAC ÷ BAC) sizing it against the budget. CV tells you where you are; VAC commits to where you'll land.

The reality test most reports omit

The **To-Complete Performance Index (TCPI)** asks: *what cost efficiency must the remaining work run at to still hit the target?* AI PMO computes two:

- **TCPI to budget = (BAC − EV) ÷ (BAC − AC)** — the efficiency the rest of the job needs to finish on the original budget.
- **TCPI to forecast = (BAC − EV) ÷ (EAC − AC)** — the efficiency needed to hit the current forecast.

The test is the comparison with current CPI. If a project runs at CPI 0.90 and the TCPI to recover the budget is 1.15, the message is blunt: *you have never performed better than 0.90, and now you'd need 1.15 for the rest of the job — the budget is gone; re-baseline the forecast and stop pretending.* A TCPI more than about 0.05 above demonstrated CPI is, in practice, not recoverable. This single comparison kills the most common reporting fiction — the perpetual "we'll claw it back next quarter."

A worked example

A project part-way through:

- **CV = −$6M (CPI 0.92)** — $6M over cost on the work done so far.
- **SV = −$4M (SPI 0.96)** — modestly behind plan.
- **VAC = −$18M (−4.5% of BAC)** — heading for an $18M overrun on current performance.
- **TCPI to budget = 1.14 vs CPI 0.92** — recovery to budget needs efficiency the project has never shown; **not recoverable.** The honest action is to adopt the forecast, re-baseline to the EAC, and manage the $18M — not to promise it back.

AI PMO owns none of these inputs. It reads the earned-value figures (BAC, PV, EV, AC), derives CV, SV, VAC and both TCPIs, then classifies recoverability by comparing TCPI against CPI — the evidence behind the Variance Analyst's commentary. It diagnoses the gap and flags when recovery is no longer credible; re-baselining and setting the forecast belong to the forecasting and change processes it triggers.

> **CV and SV size the position, VAC commits to a landing, and TCPI tests whether recovery is arithmetically possible** — moving a status conversation from optimism to evidence.

Worked figures use the synthetic demonstration portfolio.

Part IV — Communicating the Intelligence

How the numbers become legible — colour and provenance, the executive dashboard, the project workspace, and the agents.

Colour-Coding KPIs by System of Origin

Seeing at a glance where every number on the dashboard comes from

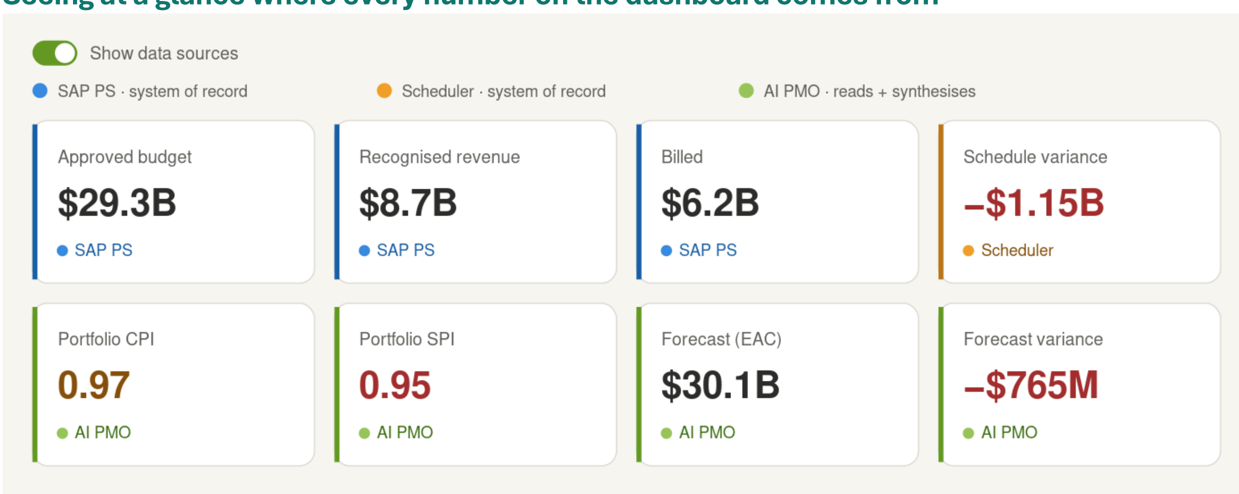


Figure 13 — The dashboard KPI ribbon with data-source colour coding — blue accents for SAP PS figures, amber for scheduler-driven figures, green for AI PMO's synthesised metrics. The number's colour still carries health: Portfolio SPI sits on a green (AI PMO) accent yet shows a red value because it is off track — the two signals coexist.

An executive reading a KPI wants to know two things at once: *is this number good or bad, and can I trust where it came from?* AI PMO answers both. It tints each KPI by the system its data originates in — SAP PS, the external scheduler, or AI PMO's own synthesis — so a viewer can see which numbers are systems-of-record facts and which AI PMO computes rather than passes through. It is a quiet provenance layer that sits alongside the existing health colours, not a replacement for them.

Why it fits

AI PMO already tags every record in its canonical model with the system it came from — the provenance behind the SAP ↔ scheduler ↔ AI PMO architecture. Colour-coding makes that invisible lineage visible on the dashboard. It reinforces the core positioning: AI PMO is an intelligence layer that reads the systems of record and synthesises the metrics they cannot produce alone — and it gives the audit-minded viewer data lineage at a glance.

Two signals, two channels

The design keeps two meanings on two separate visual channels so they never collide:

- **Health** stays on the number's colour — red for off track, amber for watch, green for on track.
- **Provenance** goes on a thin coloured left accent and a small named source tag (for example, "AI PMO").

Because the source is also written in words, colour is never the only signal — accessible and colour-blind safe — and a green "AI PMO" accent is never mistaken for a green "on-track" number. Portfolio SPI shows a green accent (AI-synthesised) with a red value (behind schedule); both read clearly, side by side.

The colour mapping

System of origin	Colour	Example KPIs
SAP PS — system of record	Blue	Approved budget, open commitment, recognised revenue, billed
Scheduler (P6 / MS Project) — system of record	Amber	% complete, schedule variance
AI PMO — reads + synthesises	Green	Portfolio CPI, SPI, forecast (EAC), forecast variance (VAC)

The flagship indices — CPI, SPI, earned value — are deliberately green even though they blend SAP cost with scheduler progress. Rather than fudge a single source, they are marked as "computed by AI PMO from the systems of record." That contrast is the whole point: green marks where AI PMO adds something neither source system produces on its own.

Interaction

A **"Show data sources" toggle** turns the provenance layer on and off — clean by default for a first glance, on when lineage matters. A small legend sits beside the KPI ribbon, and each card carries a hover tooltip naming the exact source and formula, for example *"CPI = EV ÷ AC, derived from SAP cost and scheduler progress."* The full treatment pairs a left accent bar with a named source tag; a lighter corner-dot-plus-tooltip variant keeps busy screens quiet.

The same language on a single project

The colour language extends naturally to one project — and arguably lands harder there, because a single project screen shows a richer mix of sources side by side: SAP cost, scheduler progress, AI-synthesised earned value, and AI-authored narrative all at once.

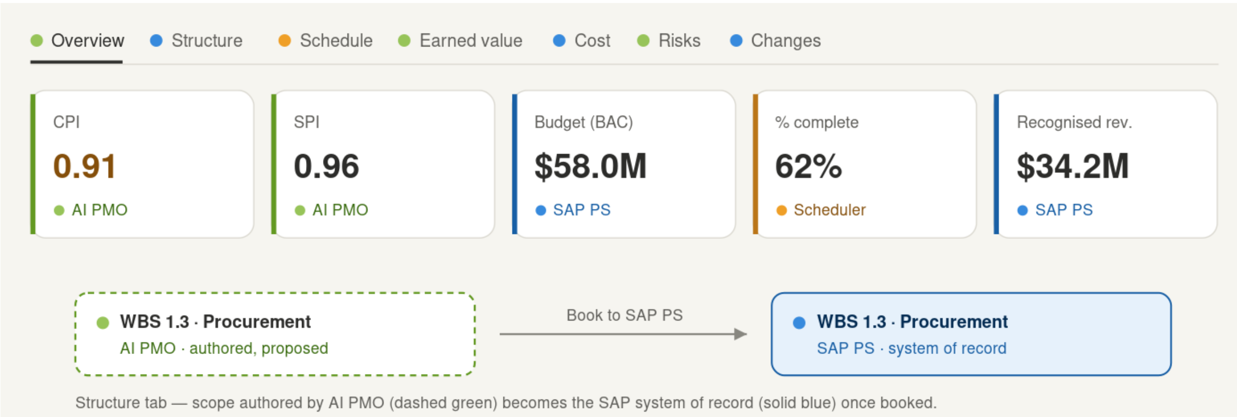


Figure 14 — A project view with data-source colour coding: each tab carries a source dot, the Overview KPI strip is colour-coded, and the Structure tab shows AI-authored scope (dashed green) becoming the SAP system of record (solid blue) once booked.

The project tabs map onto the three sources almost one-to-one:

Source	Project surfaces
SAP PS (blue)	Structure / WBS, Cost, Commitment, Billing, Results Analysis, budget & margin
Scheduler (amber)	Schedule / Gantt, % complete, resource demand
AI PMO (green)	Overview indices (CPI / SPI / EV), Variance, Risks, Issues, authored Charter & Planning narratives

The standout is the Structure tab. AI PMO authors a proposed WBS, and “Book to SAP PS” turns it into the system of record — the record’s source going from APP to SAP PS. Colour-coding makes that visible: green proposed scope literally becomes blue once booked. It is the one place provenance changes in front of the user, and the clearest single-screen demonstration of the “author, then hand to the system of record” story. Green carries two shades of meaning here — solid green for synthesised metrics (CPI / SPI / EV, derived) and a dashed outline for authored-but-not-yet-booked scope — so the two never blur.

Two guard-rails keep it coherent. The coding sits at the panel and tab grain — a source dot on each tab, an accent on each card — never per row, because tagging every cost line or task would be noise. And every surface, from dashboard to project to analytics, is driven by the same legend and one global “Show data sources” toggle, so it reads as a single visual language rather than a per-page feature.

> **Colour carries health; the accent carries provenance** — so a viewer knows in one glance both whether a number is good and how much to trust its source.

Colour & Iconography

How AI PMO makes a dense PMO legible at a glance

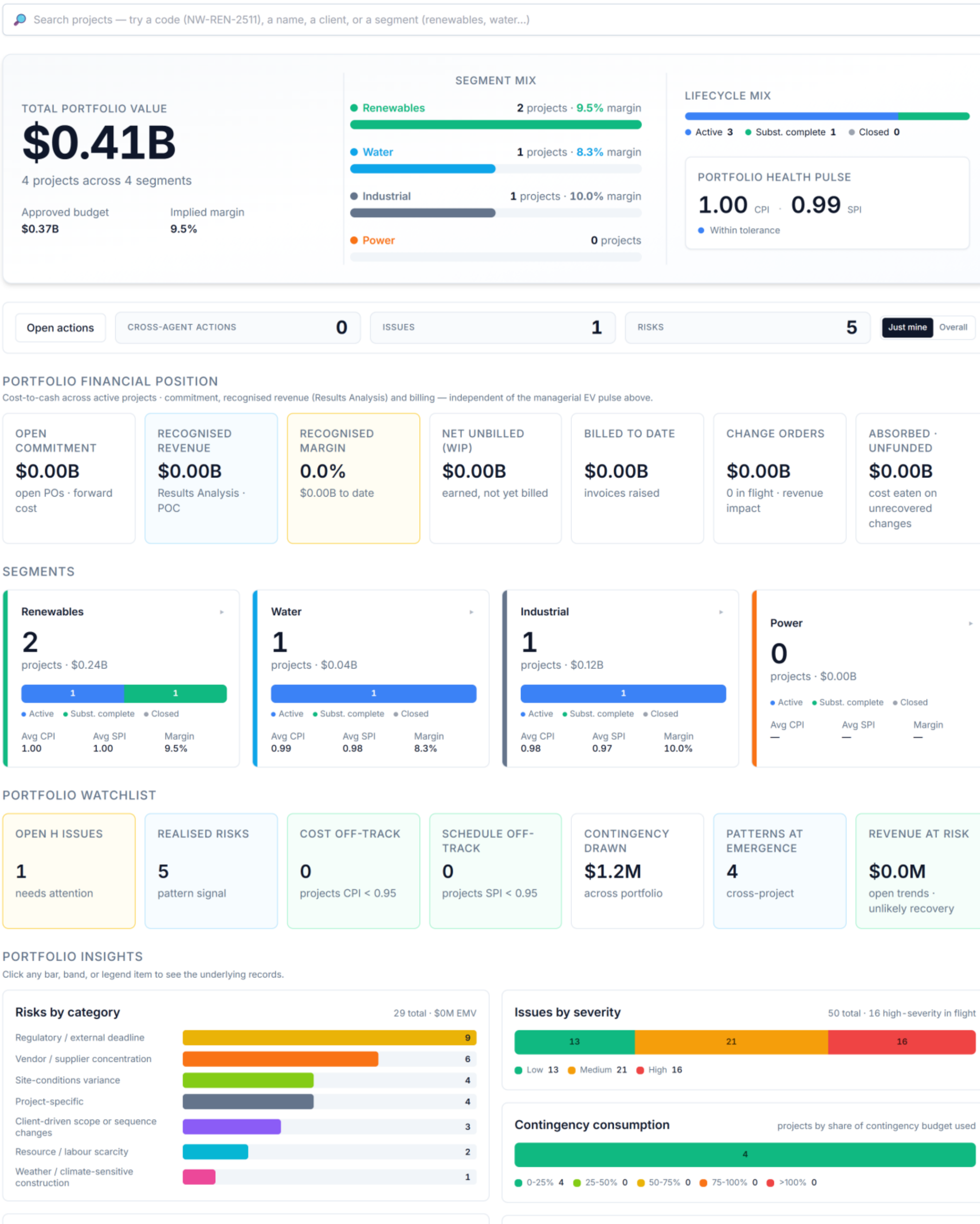


Figure 15 — The dashboard's visual language — colour by theme, icon by capability.

A portfolio dashboard can show a hundred numbers; a good one lets you *read* them without consciously decoding. AI PMO carries meaning by colour and icon, not just by labels, so a reader learns the code once and then navigates by it everywhere in the product. The failure mode in an EPC portfolio is never too little data — it is too much, undifferentiated — and a consistent visual language is the cheapest, highest-leverage way to make a complex product *feel* simple.

Colour carries meaning

Colour is never decorative; each hue maps to a concept, and that mapping holds across every screen:

- **Amber — cost.** Where the money went; cost-to-date, the cost lens.
- **Emerald — earned / revenue.** Earned value, recognised revenue, things going right.
- **Sky blue — cash.** Cash flow, funding, financing.
- **Orange — commitment.** Committed but not yet spent (open POs).
- **Red — risk / threat.** Risk exposure, breaches, things requiring attention.
- **Fuchsia — change.** Change orders and trend exposure.
- **Rose — variance.** Deviation from plan.
- **Indigo — structure.** The WBS and project structure.
- **Violet — schedule.** Time, dates, the calendar.
- **Cyan — resources.** People and capacity.
- **Blue — planning.** Plans and artefacts.
- **Slate — overview / neutral.** Summary and context.

Because the mapping is fixed, a splash of red anywhere reads as risk, amber as cost, sky as cash — before a single label is read.

Icon by capability

Every capability carries an icon drawn from the same lexicon — a receipt for cost, a rising line for earned value, waves for cash flow, a shield for risk, a gauge for performance — so tabs and cards are recognisable by shape as well as colour. On the workspace tabs the icons sit muted until a tab is active, when both the icon and its underline light up in the capability's colour; the active view announces itself in its own hue.

A caption for the question each thing answers

Alongside colour and icon, each capability earns a one-line caption stating the *question it answers*, not the feature name — on the cost lenses, for example:

- **Cost-to-date — amber, receipt** — "what we've spent."
- **Revenue recognition — emerald, rising line** — "what we've earned."
- **Cash flow — sky, waves** — "what we finance."

Colour for theme, icon for recognition, caption for the question: the pattern is what lets a non-specialist read a dense PMO screen the way a specialist does.

Provenance by colour

The same discipline extends to *where a number comes from.* A figure that is a system-of-record fact reads differently from one AI PMO has synthesised or forecast, and from one a human has overridden. Encoding provenance visually — fact versus synthesis versus forecast versus human judgement — means a reader always knows how much to trust a number, which is the foundation of the whole product's credibility.

> **A consistent visual language is a deliberate design decision, not an accident of styling** — the cheapest way to make a complex product feel simple.

Reading the Executive Dashboard

How is the portfolio, really — answered on a single screen

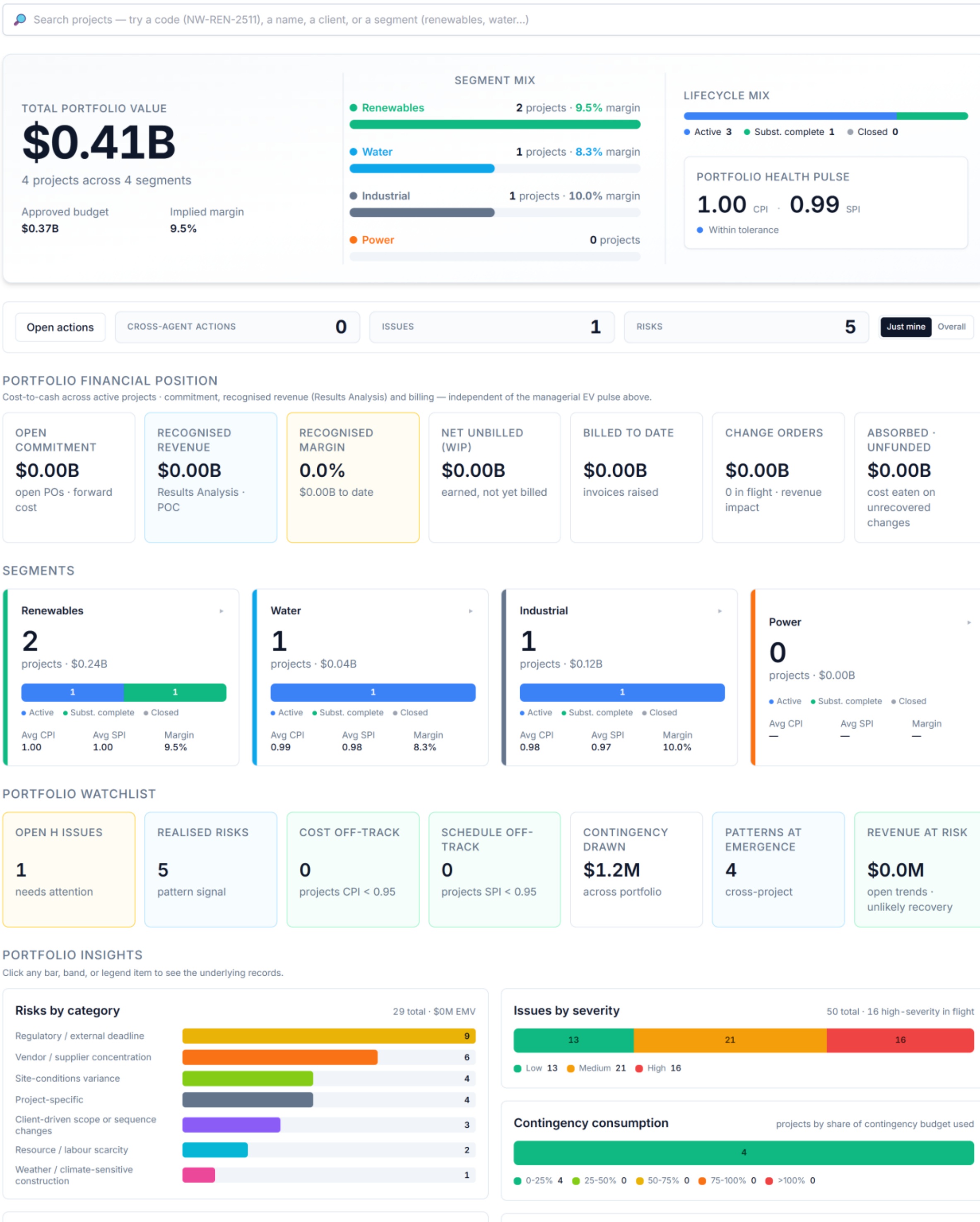


Figure 16 — The executive portfolio dashboard — read top to bottom: headline health, then the money, then what needs attention, then the patterns underneath.

The landing dashboard answers *"how is the portfolio, really?"* in one screen, laid out to be read **top to bottom as a narrowing funnel**: headline health first, then the money, then what needs attention now, then the patterns underneath, then the detail. Two rules hold everywhere. **Every KPI is clickable** — it opens a drill modal of the underlying records with a link to the full analytics page. And **colour carries meaning**, so the eye is drawn to the right hue before a label is read.

The hero

The top band is the one-glance verdict: a **headline** health statement, a **segment donut** (the portfolio's mix by segment), a **lifecycle bar** (how many projects sit in each delivery stage), and a **health pulse**. It exists to answer "should I be worried?" before any number is studied.

Portfolio earned value — the pulse

The first measured section is the portfolio **earned value** card: aggregate CPI and SPI rolled up across every active project, with the S-curve and a RAG read. This is the single most honest "on track?" number on the page — cost and schedule combined, ungameable by either alone. It is the pulse the rest of the page elaborates.

Financial position — the money

A ribbon of the cost-to-cash figures, left to right following the money:

- **Open commitment** — value of open purchase orders; forward cost not yet incurred.
- **Recognised revenue** — what may be booked to date (cost-based POC, IFRS 15).
- **Recognised margin** — recognised revenue minus cost of sales, as a percentage.
- **Net unbilled (WIP)** *or* **Deferred / over-billed** — the balance-sheet read: a contract asset when revenue is recognised ahead of billing, a contract liability when billed ahead.
- **Billed to date** — invoices raised.
- **Change orders** — total revenue impact plus how many are in flight.
- **Absorbed · unfunded** — cost eaten on changes that couldn't be recovered; margin leakage. Clicks through to the change register.

Read together, this ribbon is the portfolio's commercial position in seven numbers.

Segments

Cards for each segment — renewables, water, industrial, power — each clickable to drill into that segment's projects. This is where the work, and the exposure, concentrates.

Watchlist — what needs attention

Where the financial ribbon is the position, the watchlist is the **alarm panel**: every tile is a count or amount of something going wrong, each drilling to the records:

- **Open H issues** — open high-severity issues.
- **Realised risks** — risks that have materialised.
- **Cost off-track** — projects running CPI < 0.95.
- **Schedule off-track** — projects running SPI < 0.95.
- **Contingency drawn** — total contingency consumed across the portfolio.
- **Patterns at emergence** — cross-project signals reaching their threshold.
- **Revenue at risk** — open-trend revenue weighted by unlikely recovery; the change exposure that threatens margin.

If a manager reads only one section in a hurry, it is this one.

Insights — the patterns underneath

Five mini-charts turn the counts into distributions, each bar clickable:

- **Risks by category** — where risk concentrates by cross-cutting class, with total EMV.
- **Issues by severity** — the open issue mix, Low / Medium / High.
- **Contingency consumption** — projects banded by share of contingency used.
- **Change exposure by segment** — open-trend revenue-at-risk plus absorbed cost, by segment: *where* the exposure sits.
- **Change exposure by driver** — the same exposure by root-cause driver: *why* it is happening.

These answer “what is driving the headline?” — the diagnostic layer beneath the watchlist.

The operational tail

Below the insights sit a **hot list** of the handful of projects most needing attention (a composite of CPI/SPI deviation, open high issues and realised risks), **your recent activity** (the agent narratives you’ve generated), and a **role-specific KPI strip** — tiles chosen for the colleague’s role that are net-new versus the shared ribbons above. The page closes with the action ribbon, where cross-agent tasks are assigned and answered.

How the dashboard varies by role

The same portfolio, seen from different chairs. The shared sections — earned value, financial position, watchlist, insights — are **identical for every role**: there is one portfolio truth, not a set of per-role silos. What personalises the page is the *lens*, not the data:

- **Identity.** The header carries the colleague's role title — Senior PM, Commercial Manager, Portfolio Risk Analyst, VP Sponsor, and so on across ten seeded roles.
- **The role KPI strip.** A "Your {role} KPIs" band chosen for that remit and net-new versus the shared ribbons — the Commercial Manager sees change-order economics, the Risk Analyst sees open / high / mitigated risk counts, the HSE Manager sees HSE metrics.
- **Agent access.** Each role may invoke a curated subset of the agent roster relevant to its work, so the assistant offers the right tools rather than all of them.
- **Write capability.** Create-capable roles — PM, commercial, controls — can launch project intake, assign tasks and run agents that change state; read-only roles such as the VP Sponsor get the same visibility without the controls.
- **Section emphasis.** Each role definition names the dashboard sections most relevant to it, so the page leads with what that colleague came to see.

The same portfolio, three different chairs — note the role title in the header and the bespoke KPI strip in each:

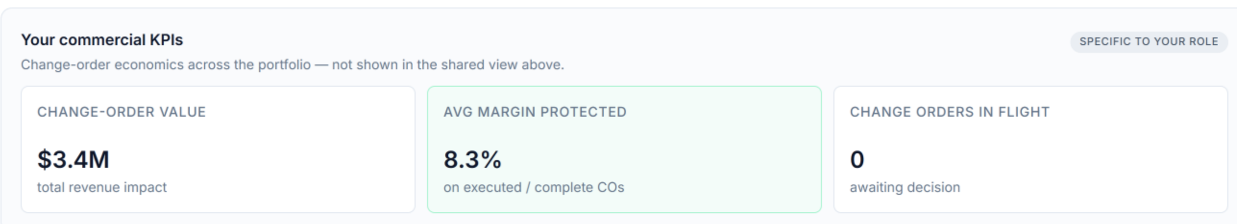


Figure 17 — The Commercial Manager's dashboard — change-order economics in the role KPI strip.

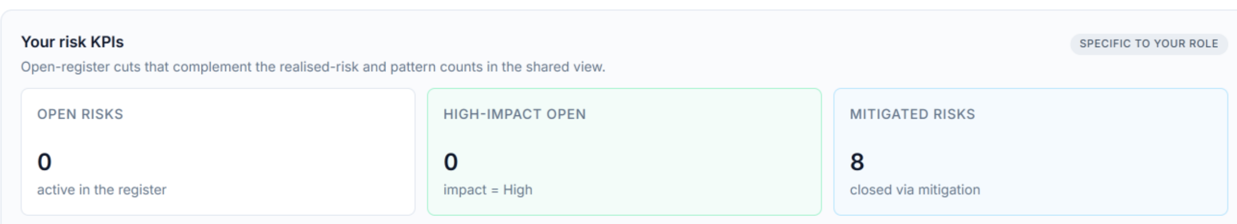


Figure 18 — The Portfolio Risk Analyst's dashboard — open / high / mitigated risk counts.

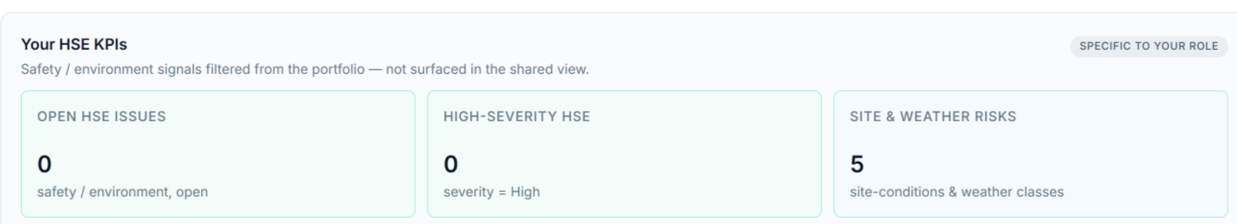


Figure 19 — The HSE Manager's dashboard — HSE-specific KPIs.

Access is conveyed by an unguessable URL token per role, and the role definition — identity, agents, sections, write permission — shapes the view, giving each colleague their relevant cut of a single, shared portfolio truth.

What makes it work

Everything above rests on two design choices: **provenance** — every figure's source, whether a system-of-record fact, AI synthesis, forecast or human override, is encoded so the reader knows how much to trust it — and the **visual language** — colour by theme, icon by capability, a one-line caption stating the question each thing answers.

> **The dashboard is not a wall of numbers; it is a single, readable argument about the health of the portfolio** — and every number on it is one click from the records behind it.

Worked figures use the synthetic demonstration portfolio.

The Analytics Pages

When “something is wrong” isn’t enough — the workbench behind the flag

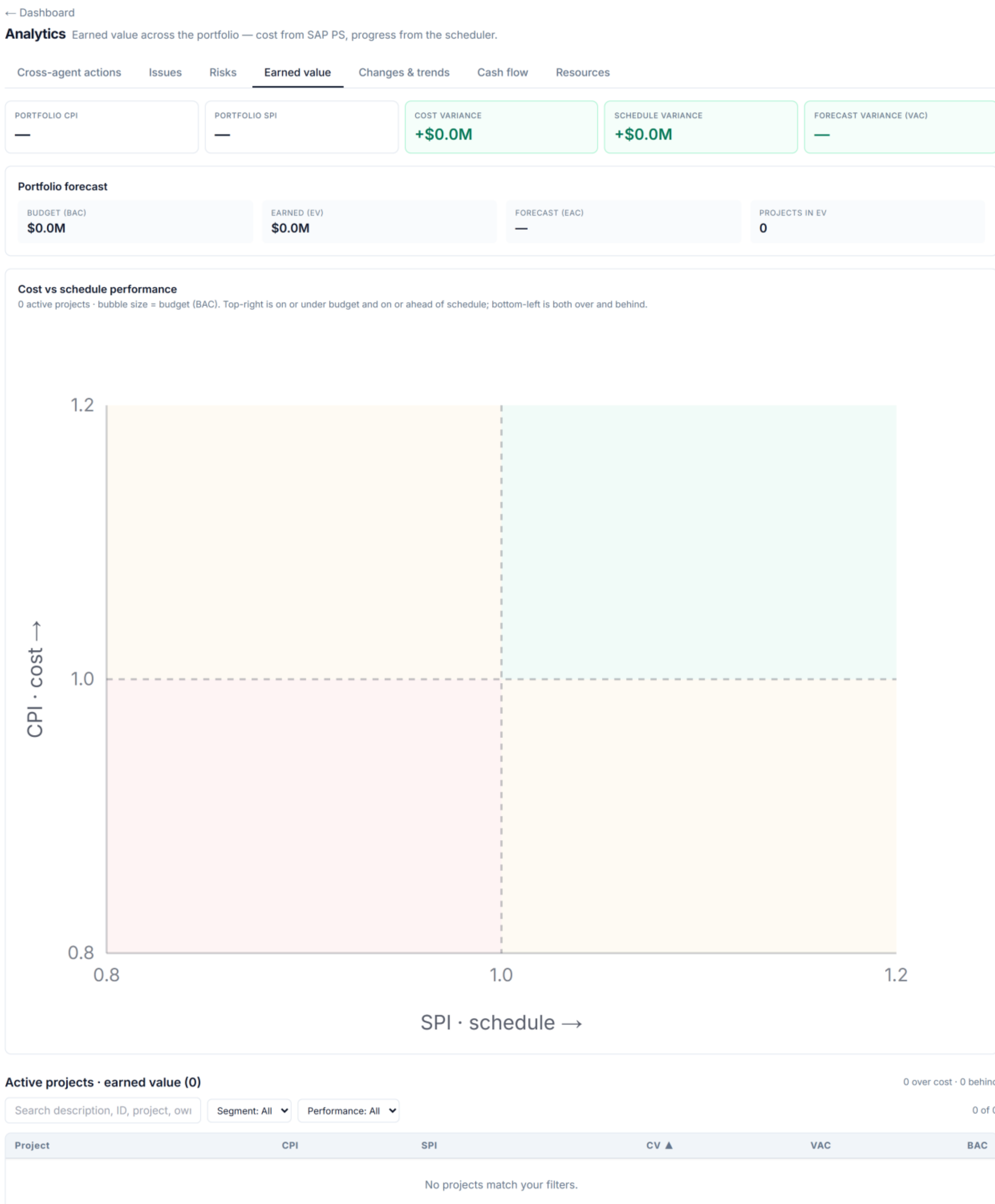


Figure 20 — Portfolio earned value analytics — the CPI×SPI quadrant and the worst-performers register.

The dashboard tells you *whether* anything is wrong. The **analytics pages** are where you go to work it — a dedicated, portfolio-wide page per domain, reached from every “see all” link and from the analytics navigation. A glance flags the problem; this is the bench where you sort, filter and act on every record behind it.

Every analytics page shares one shape: **charts on top, a register below.** The charts give the distribution and the outliers; the register is a sortable, filterable table of the underlying records across all active projects, each row linking straight to its project. One layout, seven lenses.

Earned value

The portfolio earned-value page plots every project on a **CPI × SPI quadrant** — the four corners being on-track, over-cost, behind-schedule, and both — so the projects in trouble separate themselves visually. Below it, a searchable **worst-performers register** ranks projects by cost performance. This is where a headline like “cost off-track” becomes the full league table.

Risks

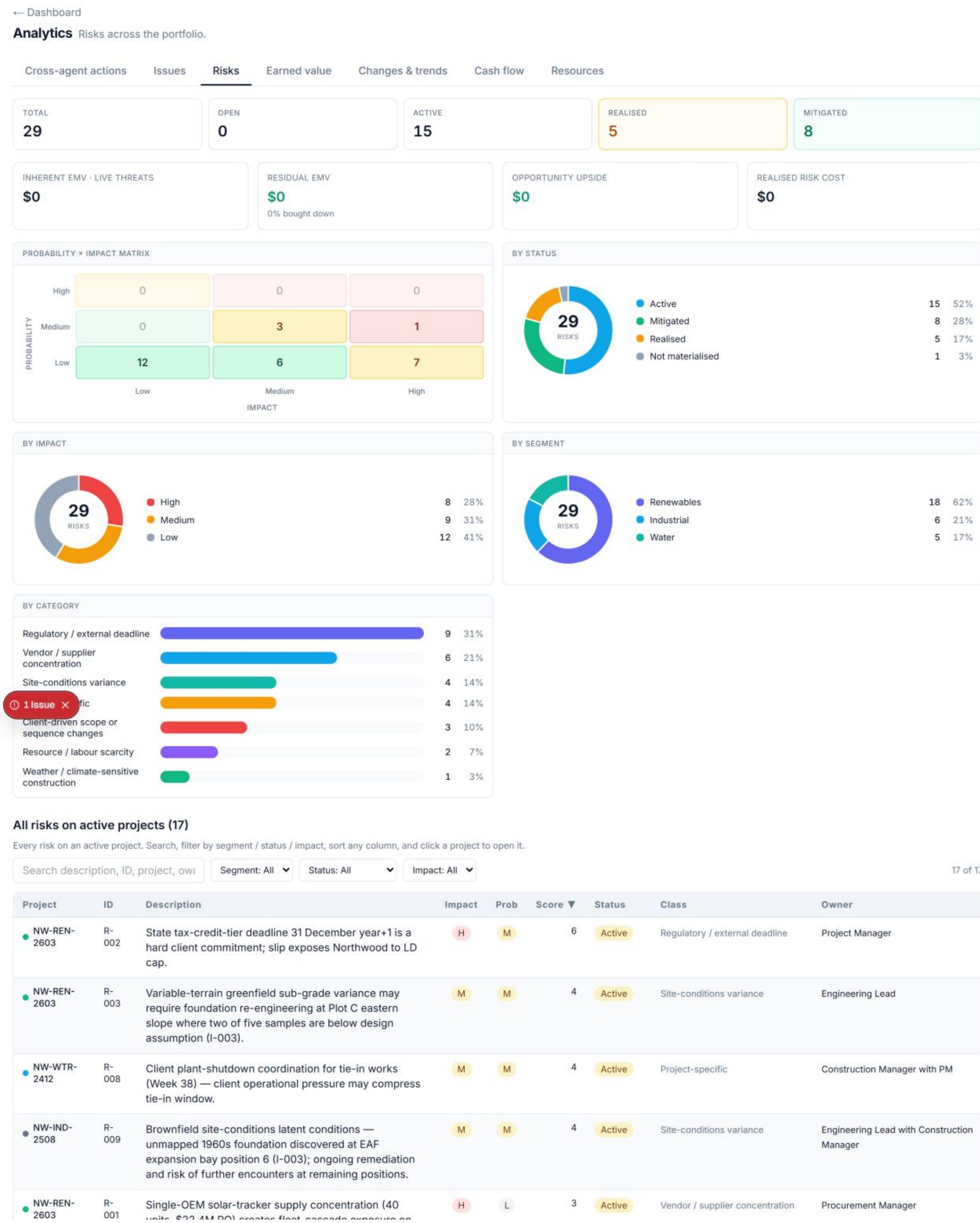


Figure 21 — Portfolio risk analytics — EMV exposure and the risk heatmap.

The risk page totals **EMV exposure** across the portfolio, shows the **inherent vs residual heatmap** (probability × impact, before and after mitigation), and lists the full risk register —

filterable by class, status and owner. It answers, at a glance, where the risk price sits and whether mitigation is buying it down.

Changes

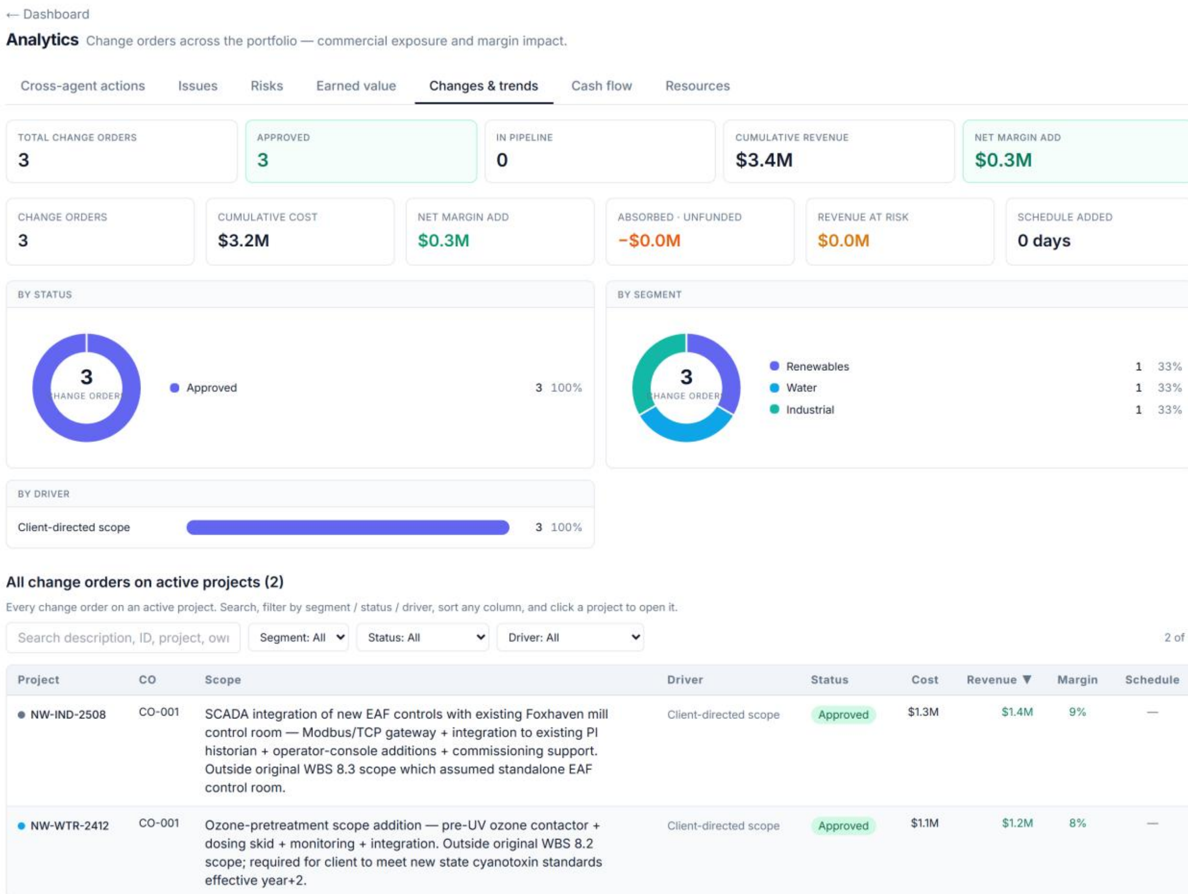


Figure 22 — Portfolio change & trend — funded vs absorbed, with revenue-at-risk.

The changes page is the portfolio trend register: KPIs for total changes, executed and in-pipeline, cumulative revenue and net margin, plus the exposure pair — **Absorbed (unfunded)** and **Revenue-at-risk**. Donuts break changes down by status and segment, a bar ranks them by driver, and the register lists every change order.

Cash flow

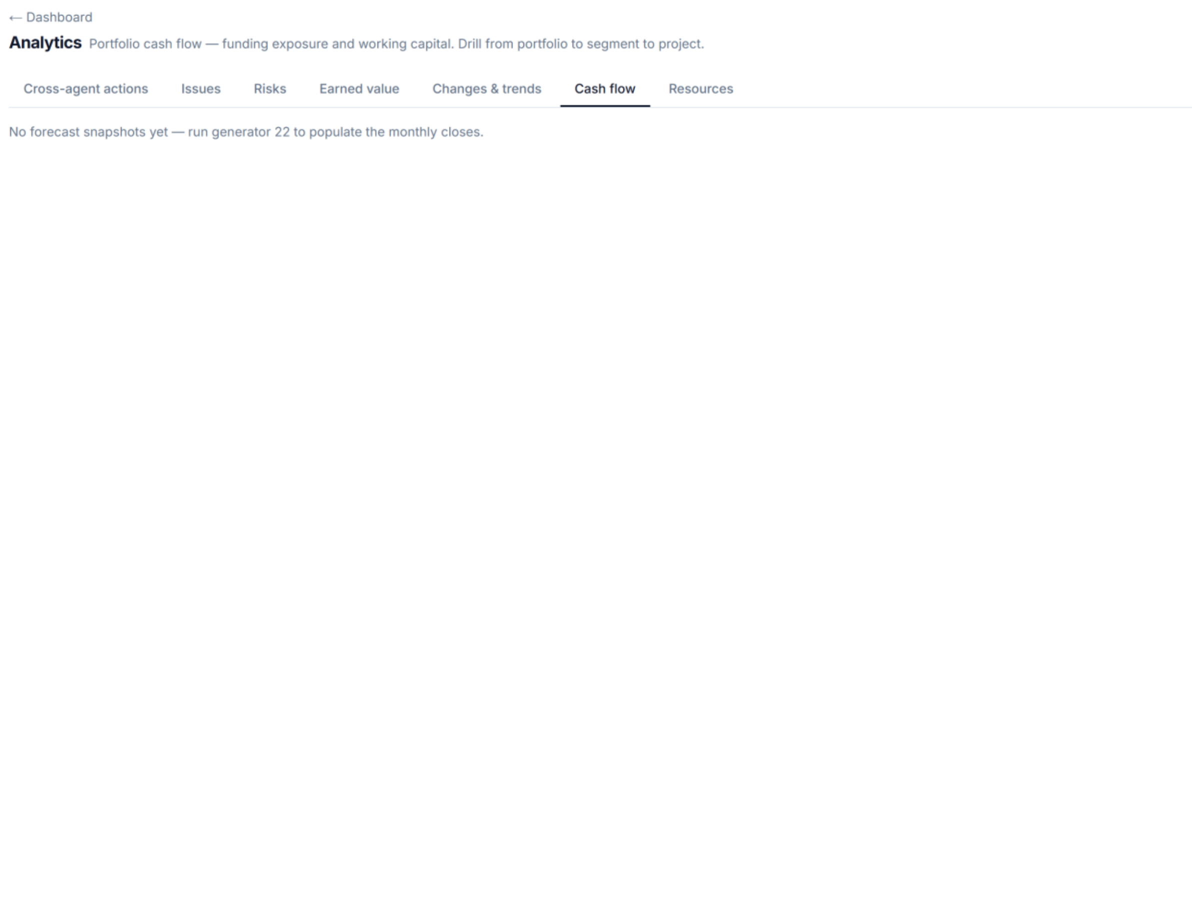


Figure 23 — Portfolio funding exposure with drill to segment and project.

The cash-flow page is the portfolio funding-exposure explorer: the whole book's cash curve — total working capital, peak funding requirement and cash-positive crossover — with **drill-down from portfolio to segment to a single project**. It answers the treasurer's question at every level of zoom.

Issues, resources and actions

Three further pages complete the set:

- **Issues** — portfolio issue health (aging, SLA breaches, severity mix) and the full issue register.
- **Resources** — resource demand rolled up by role and segment, with segment drill-down: who is needed where.
- **Actions** — open and closed assignments across the portfolio, making the assign-to-respond loop visible.

Why the pages matter

The dashboard is for *noticing*; the analytics pages are for *working*. Together they give a portfolio lead the two modes the job actually needs — a glance that flags the problem and a workbench that sorts, filters and acts on every record behind it, without ever dropping down to a spreadsheet. Every page reads the full portfolio through paginated queries, so no project is ever silently left off the table.

> **The dashboard says what to look at; the analytics pages let you do something about it** — noticing and working, kept one click apart.

The Project Workspace

One page that answers "how is this job?"

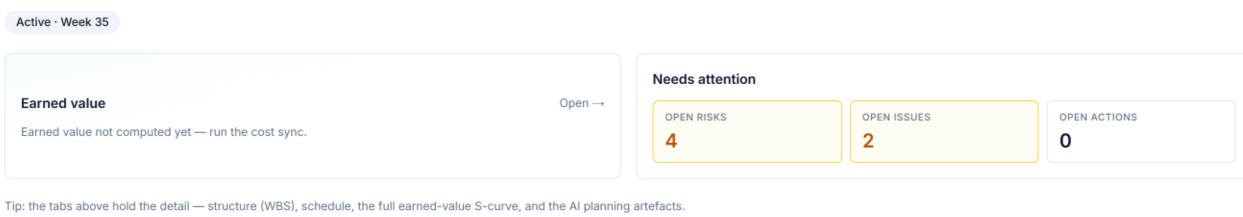


Figure 24 — The project Overview — earned value, what needs attention, the margin bridge and the schedule envelope on one screen.

Click into any project — from a dashboard drill, an analytics register, or the global search — and you land in the **project workspace**: the single-project counterpart to the portfolio view. Where the dashboard answers *"how is the book?"*, the workspace answers *"how is this one job?"* — through a single tabbed page, colour-coded by capability, with a context-aware agent assistant floating alongside that already knows which project you are looking at.

The tabs follow the project's own logic — structure, then time, then money, then the registers, then the plan.

Overview — the one-screen verdict

The landing tab leads with what matters: the project's **earned value** (CPI/SPI), a **needs-attention** panel (open risks, high issues, actions awaiting response), the **margin bridge** (as-sold to as-built), and the **schedule envelope** (forecast finish against the contract date). It is the project's health in one view.

Structure — the WBS

Figure 25 — The Structure tab — the canonical WBS with provenance.

The canonical **work breakdown structure**: the colour-coded WBS tree that is the join key between cost and schedule, with provenance showing which branches are booked to the system of record and which were AI-authored and approved. Risks and issues are shown mapped to their WBS branch.

Schedule

The project timeline and milestones — the scheduler's contribution, framed as a forecast-versus-contract reconciliation rather than a re-scheduling.

Earned value

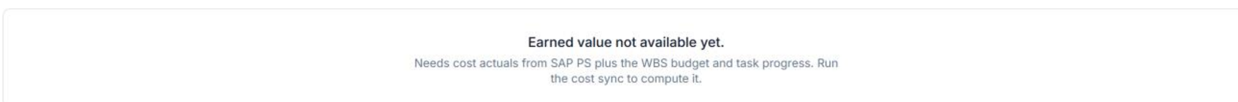


Figure 26 — The Earned value tab — S-curve, indices, earned schedule, EV by WBS and the forecast trend.

The flagship tab: the **S-curve** (PV/EV/AC), CPI/SPI and earned schedule, **EV by WBS branch** (which branch carries the overrun), and the **forecast-trend** chart with the margin-at-complete band.

Cost

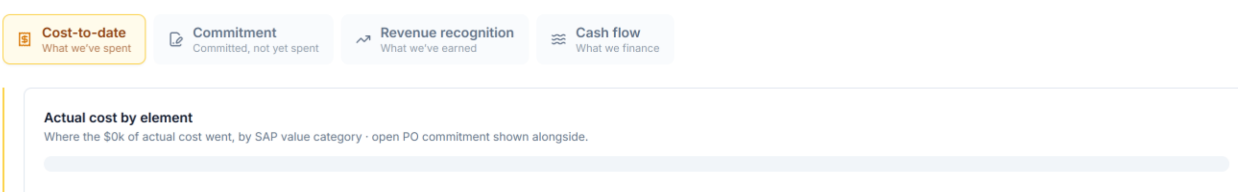


Figure 27 — The Cost tab — four lenses behind one segmented control.

A single financial tab with four lenses behind a segmented control, telling the money story in order: **Cost-to-date** (where it went, by element and WBS phase), **Commitment** (open POs, committed but not spent), **Revenue recognition** (cost-based POC, WIP and deferred), and **Cash flow** (funding exposure).

Risks & issues, Changes, Variance

The registers and their synthesis:

- **Risks & issues** — the project risk register (inherent/residual EMV, mitigation) and the issue queue (aging, SLA, priority).
- **Changes** — the change & trend register: pipeline, margin impact, drivers, funded-versus-absorbed and recovery confidence.
- **Variance** — CV/SV, VAC and the TCPI recovery test.

Planning

The AI-authored planning artefacts for the project — charter, schedule analysis, budget basis, communications and the rest — each editable with provenance, produced by the planning agents.

Why a single workspace

Before AI PMO, answering "how is this project?" meant opening SAP for cost, the scheduler for time, and a stack of spreadsheets for the registers. The workspace collapses that into one navigable page where every tab is the same project seen through a different capability — cost and schedule already reconciled on the WBS, the registers quantified, and an assistant on hand to narrate any of it.

> **The workspace applies the portfolio's logic one level down** — one synthesis, many lenses, no spreadsheet.

The AI Agents — Your Specialist Team

AI PMO is not a single chatbot. It is a **roster of specialist agents** — fifteen of them — each grounded in a recognised project-management method and each responsible for a narrow, well-understood job. One drafts a charter; another reasons about the critical path; another controls the cost-to-cash position; another reviews a change order four ways for commercial exposure. You never have to remember which is which: you type what you need in plain language, and the system routes you to the right specialist.

This chapter covers what the agents are, how you work with them, what they can read and write, and which agents each role is allowed to use — the governance that keeps a powerful capability safe.

One surface for everything: the Ask AI Assistant

Every interaction happens in one place — the **Ask AI Assistant**, the floating panel in the bottom-right of every page. There is deliberately no second console, no separate "create a risk" form, no admin screen. Whether you are *asking* a question, *raising* a risk, or *assigning* a task to a colleague, you do it by typing into the same assistant. That is the single most important design decision in the product: **one surface, so there is one place to learn, one place to govern, and one audit trail.**

The assistant is **context-aware**. Open it on a project page and it works against that project's data; open it on the portfolio dashboard and it works across the whole book. The header always tells you which. When the panel is empty it offers **role-aware starter prompts** — a "Create / Update / Assign" row and an "Ask agent" row — so the actions available to you are discoverable without a manual. Pick one, edit it, send.

By default the assistant is set to **Auto**: you type, and a lightweight router reads your request and picks the right specialist. Power users can override with the agent dropdown, but most of the time you should not have to think about which agent you need — that is the router's job.

The roster

Each agent does a few things well and deliberately *not* others, so it never strays outside its competence. The table is the quick reference; ask the assistant the example prompt to see any agent in action.

Agent	What it answers	Try asking
Charter Drafter	The founding charter — scope, deliverables, milestones, governance	"Draft the charter for this project."
Stakeholder Analyst	Who has a stake, and how to engage each	"Build the stakeholder register."
WBS Builder	The work broken into deliverable packages	"Build the WBS to Level 1–3."
Schedule Reasoner	Critical path, float, sequencing risk	"What's the critical path, and what's most at risk?"
Cost Planner	Approved/quoted cost total allocated across categories and time-phased	"Allocate the approved budget across the major cost categories."
Communications Planner	Who hears what, how often, through which channel	"Build the comms plan."
Issue Logger	What open issues matter most, and why	"Summarise the open issues; flag overdue high-severity ones."
Variance Analyst	Are we on cost and on schedule? (CPI/SPI)	"Summarise the variance position and flag concerns."

Agent	What it answers	Try asking
Change Order Reviewer	Is a change order commercially sound?	"Review CO-U01 — is margin protected?"
Risk Analyst	The project risk register, made current and readable	"What are the top risks to watch, and why?"
Lessons-Learned Synthesiser	The lessons worth adopting firm-wide	"What are the top three firm-level lessons here?"
Closeout Reporter	How the project actually went, vs baseline	"Draft the closeout executive summary."
Portfolio Risk Reviewer	Risk patterns that span *many* projects	"Where is cross-cutting risk emerging across the portfolio?"
Status Reporter	The one-page weekly status, tailored to the reader	"Write this week's status report for the sponsor."
Cost Controller	Commitment, cost-to-date, and earned-vs-billed	"Give me the cost and commitment position."

Every agent is grounded in a standard — PMBOK knowledge areas for the planning and control agents, Earned Value Management for variance, a four-frame commercial analysis for change orders, IFRS-15 and results-analysis for revenue. The methods are not improvised; they are the reason the answers are defensible.

Two ways an agent helps: reading and writing

Most of what you ask is **read-and-synthesise**: summarise the variance, surface the top risks, explain where the critical path runs. The agent reads the project's data and gives you a grounded answer — fast, between the formal monthly reviews.

But the agents can also **write** — always through the same chat, never a separate form:

- **Raise an entry.** Type "log a vendor risk: the sole-source transformer supplier may slip delivery by six weeks" and the Risk Analyst drafts a structured risk and shows you an editable **confirm card** in the chat. Review the inferred fields, correct anything, click **Add to register**. The same flow raises **issues** and **change / trend entries**.
- **Assign a task.** Type "assign a task to Procurement: pre-qualify a second transformer supplier — high urgency" and the agent drafts it into an **Assign actions** card; confirm, and it lands in that colleague's action queue.

Two principles govern every write. First, **human in the loop**: the agent never writes silently — it *drafts*, you *confirm*, and the confirm card is always the last step, so nothing reaches a register or a colleague's queue without your explicit click. Second, **provenance is honest**: AI PMO owns the things no ERP does, so the risk register and the issue log live here, and a risk or issue you raise through the agent is simply **agent-raised** in AI PMO's own register. A **change order** is different — SAP PS is its system of record, so a change you raise is **provisional until it is booked into SAP PS**, and it round-trips out through the existing export, clearly marked until then. Either way, everything you raise is tagged so its origin is never in doubt.

Who can do what — agents by role

Not every role can use every agent. Each role is granted only the agents that fit its remit — a least-privilege model — so the starter prompts and the create/assign actions a person sees are exactly the ones they are allowed to use. A read-only role sees no create or assign actions at all.

The table shows the **create and assign** capabilities by role. (The read-and-analyse asks adapt the same way — Procurement sees the change-order ask, the Cost Controller sees the cost-and-commitment ask, and so on.)

Role	Raise risk	Log issue	Change / trend	Assign task
Senior PM (PMO Director)	✓	✓	✓	✓
Program Manager	✓	✓	✓	✓
Risk Analyst	✓	✓	—	✓
Procurement Strategist	✓	—	✓	✓
Commercial Manager	—	—	✓	✓
Construction Manager	—	✓	✓	✓
Engineering Manager	✓	—	—	✓
HSE Manager	—	✓	—	✓
Project Controls Manager	—	—	—	✓
VP Sponsor	—	—	—	—

Each cell follows directly from the agents the role holds: raising a **risk** needs the Risk Analyst, logging an **issue** needs the Issue Logger, a **change / trend** entry needs the Change Order Reviewer, and **assigning** a task needs write access. The **VP Sponsor** is read-only by design — an executive who consumes status, variance and patterns but does not edit state

— so the Sponsor sees analysis prompts only. The same gate is enforced on the server, not just hidden in the interface, so the permission is real.

Where these work: project vs portfolio

The table above answers *what* a role may do. *Where* it can do it depends on the context the assistant is in, and the assistant adapts its starter prompts to match — the same role sees one set of chips on a project page and a different set on the portfolio dashboard.

The rule is simple: **a risk, issue or change entry is always raised against a specific project**, so those actions are offered only when you are inside one. Assigning a task and asking for analysis work in both places — the analysis simply widens from a single project to the whole portfolio.

What you do	On a project page	On the portfolio dashboard
Raise a risk / issue / change	Yes — attached to that project	Open a project first (entries belong to a project)
Assign a task	Yes — scoped to the project	Yes — portfolio-level
Ask / analyse	*This* project: variance, top risks, cost position	*Across* projects: which need attention, where risk is emerging

> **Fifteen specialists, one surface, and a permission gate enforced on the server** — a powerful capability made safe by governing where it lives and who may act.

Part V — Justification & Governance

Why these formulas: the standards behind every number, so each figure survives an audit.

Why These Formulas

“Did you just invent these numbers?” — no, and here is the proof

The fair question anyone evaluating AI PMO eventually asks is *where do these methods come from — did you make them up?* The answer, deliberately, is no. Every calculation in the money story and the risk chapters is a recognised industry or accounting standard, applied faithfully. Using standards rather than house rules is what makes the numbers **defensible** — consistent across projects, familiar to any reviewer, and reconcilable to the systems of record. This chapter names the sources.

Earned value — EVM, ANSI/EIA-748 and the AACE practices

The earned-value method and its variances are the established **Earned Value Management** discipline, codified in the United States as **ANSI/EIA-748** and embedded in **PMI’s PMBOK Guide**. PV, EV, AC, CPI, SPI, CV, SV, VAC and TCPI are not AI PMO inventions — they are the standard EVM formulae, used verbatim. **AACE International** (the Association for the Advancement of Cost Engineering) publishes the Recommended Practices governing how the inputs — progress measurement, estimate classes, forecasting — should be prepared. AI PMO’s contribution is the *synthesis and timeliness*, not the formulae.

Forecasting to EAC — AACE forecasting practice

The three EAC methods — budget-at-completion over CPI; actuals plus remaining budget; and the CPI×SPI-weighted forecast — are the standard EVM forecasting formulae, with the choice between them guided by AACE’s forecasting recommended practice and the TCPI reality test. Picking a forecast is a judgement; the methods that bound it are standard.

Revenue recognition — IFRS 15

Revenue recognition follows **IFRS 15, “Revenue from Contracts with Customers”** — the governing accounting standard — applied as over-time recognition by a cost-based input method (percentage-of-completion). The WIP/contract-asset and deferred-revenue/contract-liability treatment is IFRS 15's, mirrored in SAP's Results Analysis. This is the one area where the standard is not optional: it is the basis the auditor will require.

Risk — EMV and contingency confidence

The risk chapter uses **Expected Monetary Value** (probability × impact) and inherent-to-residual reduction — the standard quantitative risk-management technique found in the PMBOK risk knowledge area — and sizes contingency against **P50/P80** confidence levels, the convention used in quantitative risk analysis.

Change and cash — contract and treasury convention

Change and trend management follows standard project-controls trend practice and reflects IFRS 15's variable-consideration constraint in its recovery-confidence treatment. Cash flow uses ordinary treasury convention — payment and collection lags, retention — made explicit as adjustable assumptions.

Why standards, not house rules

Three reasons, each load-bearing. **Defensibility:** a reviewer who knows EVM or IFRS 15 can check AI PMO's numbers against a standard they already trust. **Consistency:** the same method applied to every project makes cross-project comparison valid. **Reconciliation:** standard methods tie back to the systems of record, so the synthesis layer never drifts away from the ledger and the schedule.

> **The innovation is not in the formulae — it is in computing the standard formulae continuously, across the seam, with a narrative on top** — which is exactly why the numbers hold up under review.

Appendices

Reference material.

Glossary

EPC, earned-value and financial terms used in this book

AC — Actual Cost. The cost actually incurred for the work performed, from the cost system (SAP PS).

As-built margin. Forecast margin at completion: current contract value minus estimate at completion (EAC).

As-planned margin. Margin in the current plan: current contract value minus current WBS budget.

As-sold margin. Margin at booking: sold contract value minus the frozen baseline budget.

BAC — Budget at Completion. The total approved project budget.

CPI — Cost Performance Index. EV ÷ AC. Above 1.0 = under cost; below = over.

CV — Cost Variance. EV − AC. Negative = over cost.

Contract asset (WIP). Revenue recognised in excess of amounts billed — earned but not yet invoiced.

Contract liability (deferred revenue). Amounts billed in excess of revenue recognised — invoiced ahead of the work earned.

Contingency. Budget set aside to cover identified risk; sized against residual exposure at a P50/P80 confidence level.

EAC — Estimate at Completion. The forecast total cost of the project.

EMV — Expected Monetary Value. Probability × impact; the risk-weighted cost of a risk.

EV — Earned Value. The budgeted cost of the work actually performed; physical progress priced at budget.

Earned schedule. Schedule performance measured in time units rather than cost, correcting SPI's end-of-project drift.

EVM — Earned Value Management. The standard cost/schedule performance discipline (ANSI/EIA-748, PMBOK).

IFRS 15. The accounting standard governing revenue recognition from customer contracts.

Inherent / residual EMV. Risk-weighted exposure before / after mitigation.

MTTR — Mean Time To Resolve. Average time from an issue being raised to closed.

Peak funding requirement. The maximum cash a project ties up at once — the trough of the net cash position.

POC — Percentage of Completion. Progress measure; cost-based POC = actual cost ÷ expected cost, used for revenue recognition.

PV — Planned Value. The budgeted cost of the work planned to be done by now; the baseline.

Recognised revenue. Revenue that may be booked to date: POC × contract value (IFRS 15).

Residual exposure. Risk-weighted cost remaining after mitigation, for live risks only.

Retention. A percentage of each payment withheld by the client until completion or end of the defects period.

SPI — Schedule Performance Index. EV ÷ PV. Above 1.0 = ahead; below = behind.

SV — Schedule Variance. EV − PV. Negative = behind schedule.

TCPI — To-Complete Performance Index. The cost efficiency the remaining work must achieve to hit a target (budget or forecast); the recovery reality test.

Trend. A potential change identified before it becomes a funded change order or an absorbed (unfunded) cost.

VAC — Variance at Completion. BAC − EAC; the forecast total over- or under-run.

WBS — Work Breakdown Structure. The hierarchical breakdown of project scope; the code on which cost (SAP) and schedule (P6/MSP) are joined.

WIP — Work in Progress. See *Contract asset*.

About this edition

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